High Frequency Data Modelling: a Case Study

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Statistics and Research Department
Tax Agency

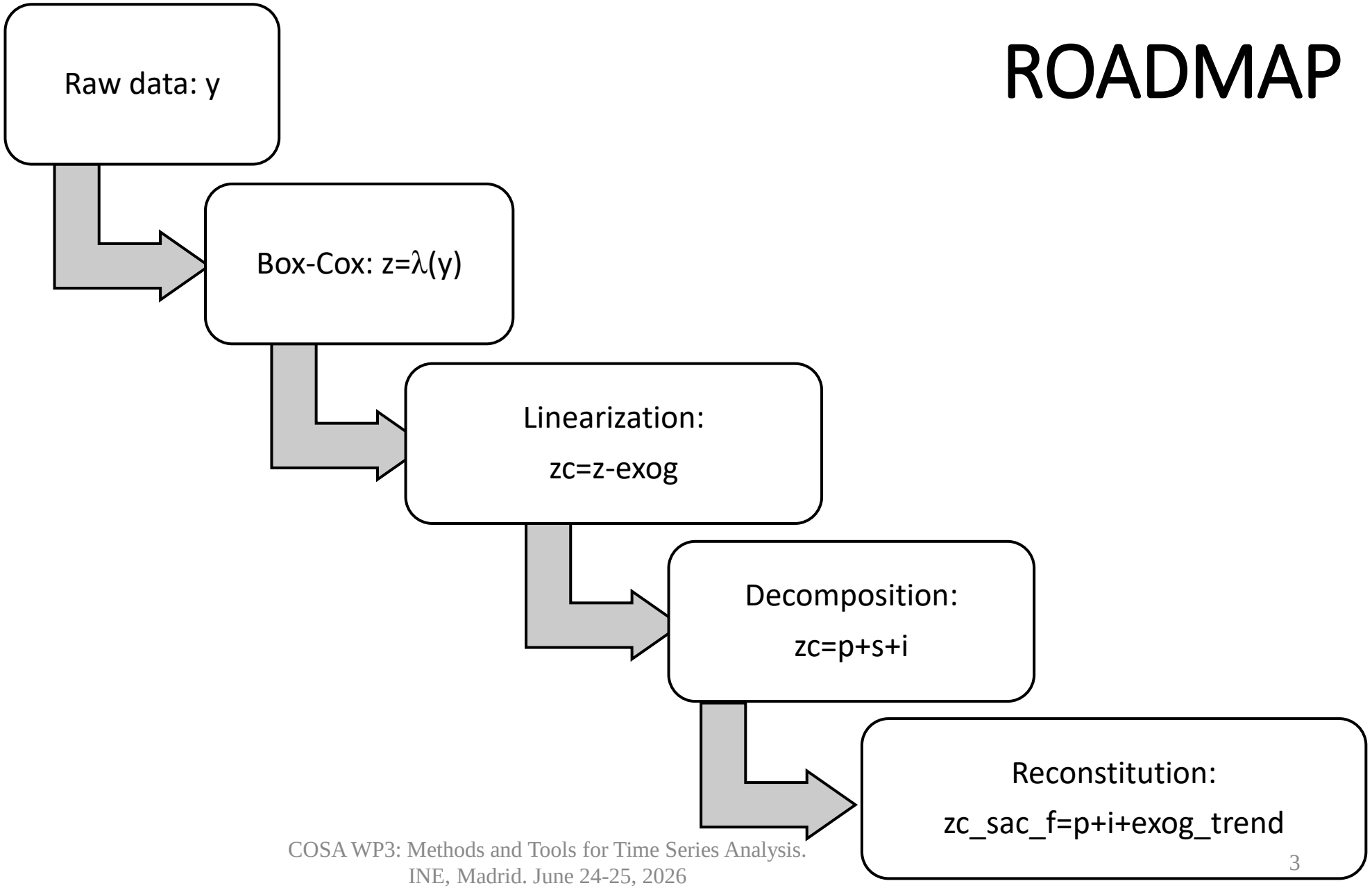
ACKNOWLEDGEMENTS AND DISCLAIMER

- We thank R. Frutos, D. Ollech and A. Smyk for their input at different stages of this project.

- The opinions are those of the authors and do not necessarily reflect the views of the Tax Agency.

- Work in progress**.** Comments and suggestions are welcome!

ROADMAP


Raw data: y
Box-Cox: z=λ(y)
Linearization:
zc=z-exog
Decomposition:
zc=p+s+i
Reconstitution:
zc_sac_f=p+i+exog_trend

INTRODUCTION

- Main purpose of the talk: a complete application of the AMB methodology embedded in the `rjd3hifreq` R package to a real (and somehow difficult) daily economic time series.

- Preliminary review of the main steps of the methodology, including some possible complements and extensions.

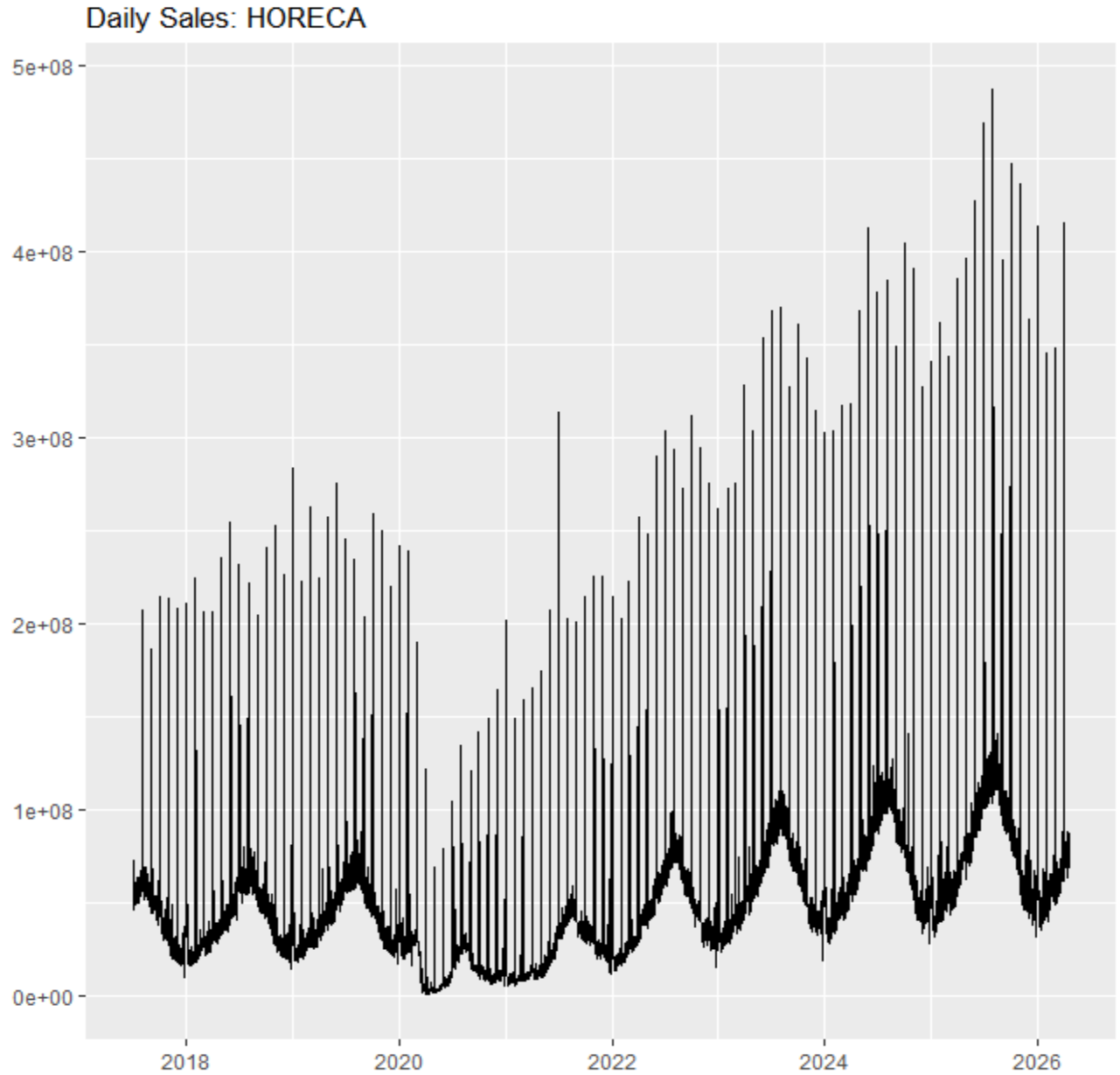
- Work-in-progress: your comments are welcome at any time.

INTRODUCTION

- A real (and somehow difficult) daily economic time series: Spanish daily sales of **HO**tels, **RE**staurants and **CA**fes (HORECA). NACE-25 Groups: 5510 – 5630.

- Compiled from the registered VAT forms by the Tax Agency. Valuation: current prices.

- Full sample: July 1st 2017 – April 21th 2026 (n=3217 ~ 8.7 years).

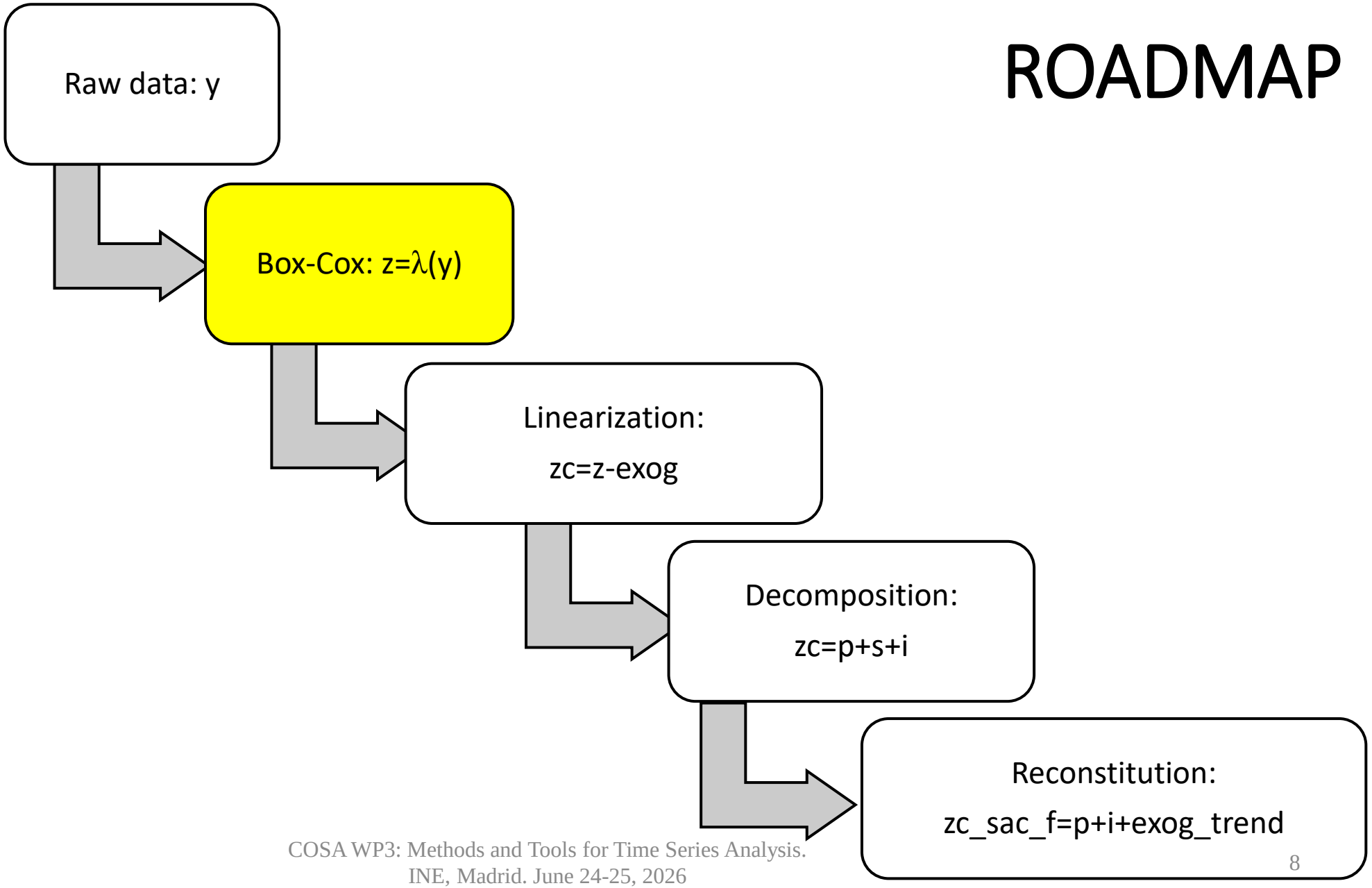

Daily Sales: HORECA
5e+08
4e+08
3e+08
2e+08
1e+08
0e+00
2018
2020
2022
2024
2026

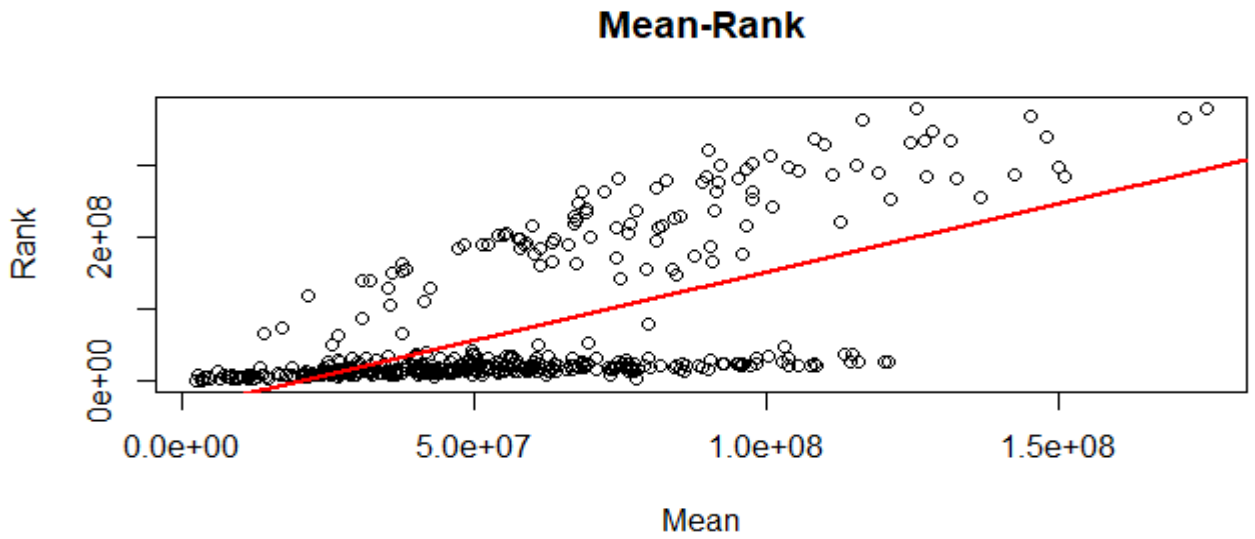
INTRODUCTION

- A daily time series full of interesting features:
  - Multiple seasonal patterns: weekly, monthly and annual.
  - Fractional seasonal patterns: monthly (30.4375d) and annual (365.2425d).
  - Calendar effects: holidays, Easter.
  - Extremely affected by the CV19 shock.
  - Outliers.

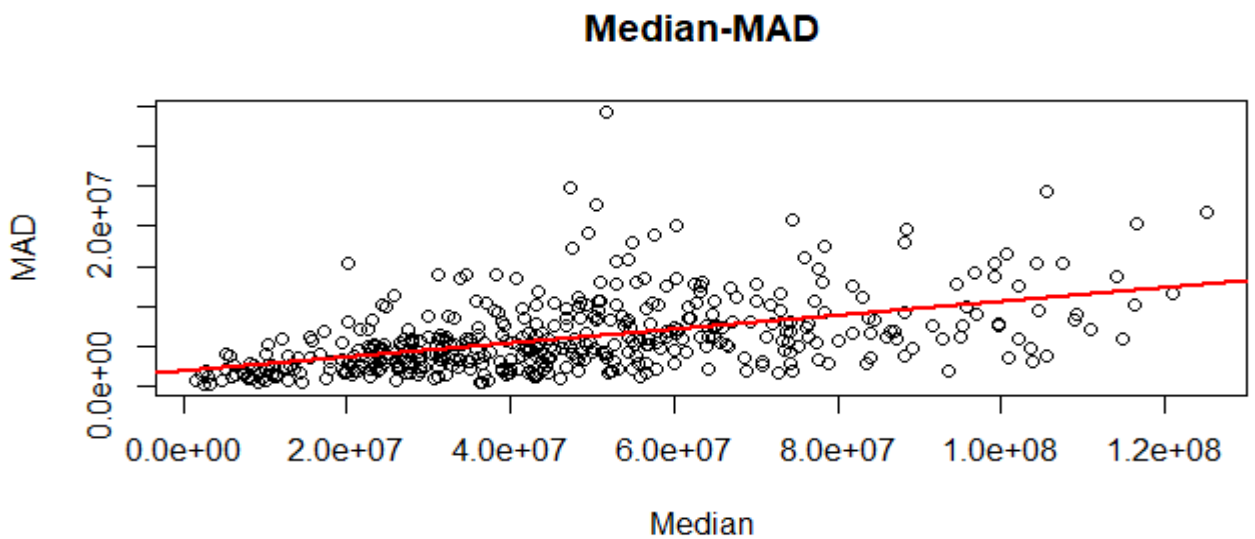
- A big mess!

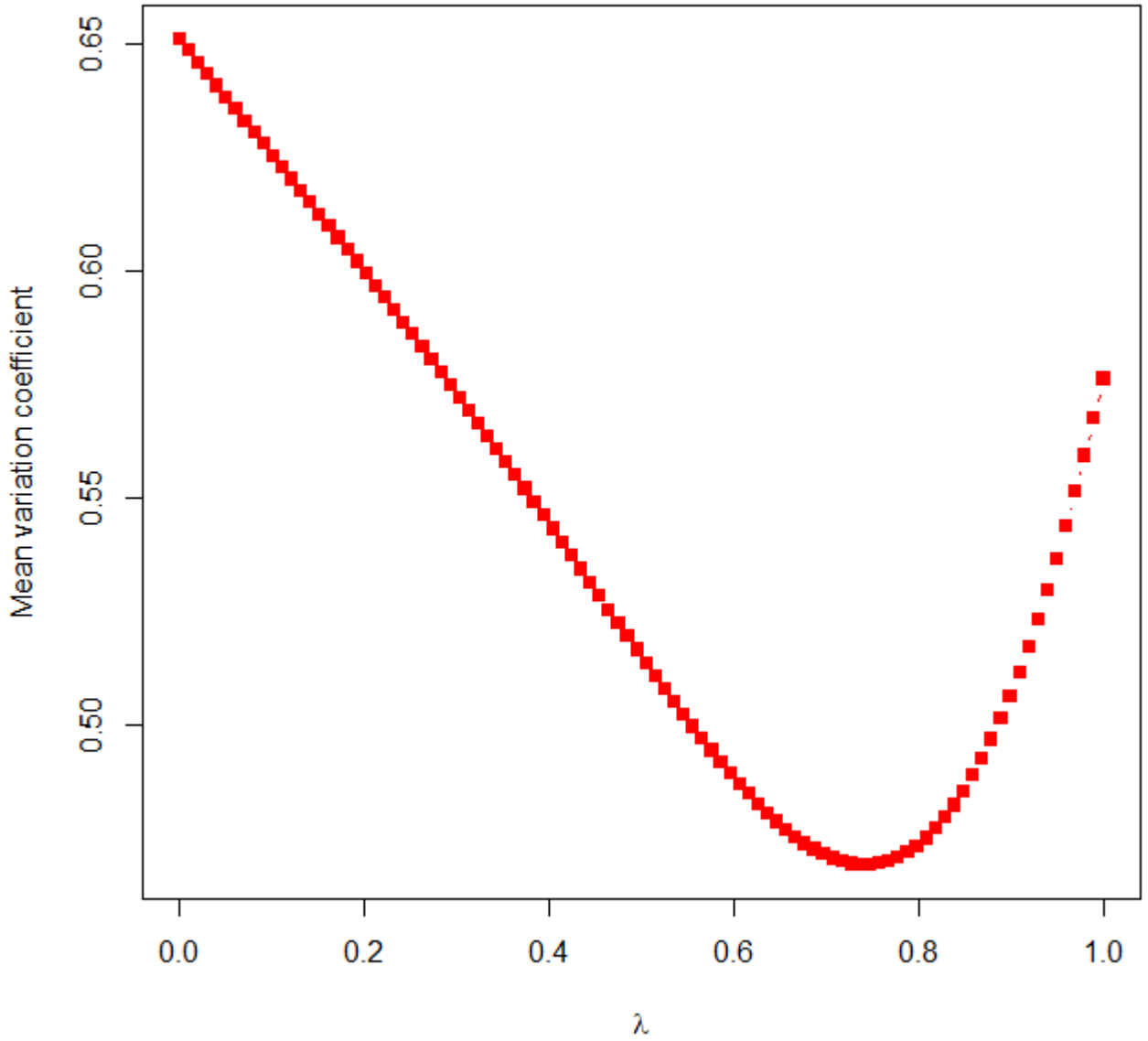
ROADMAP


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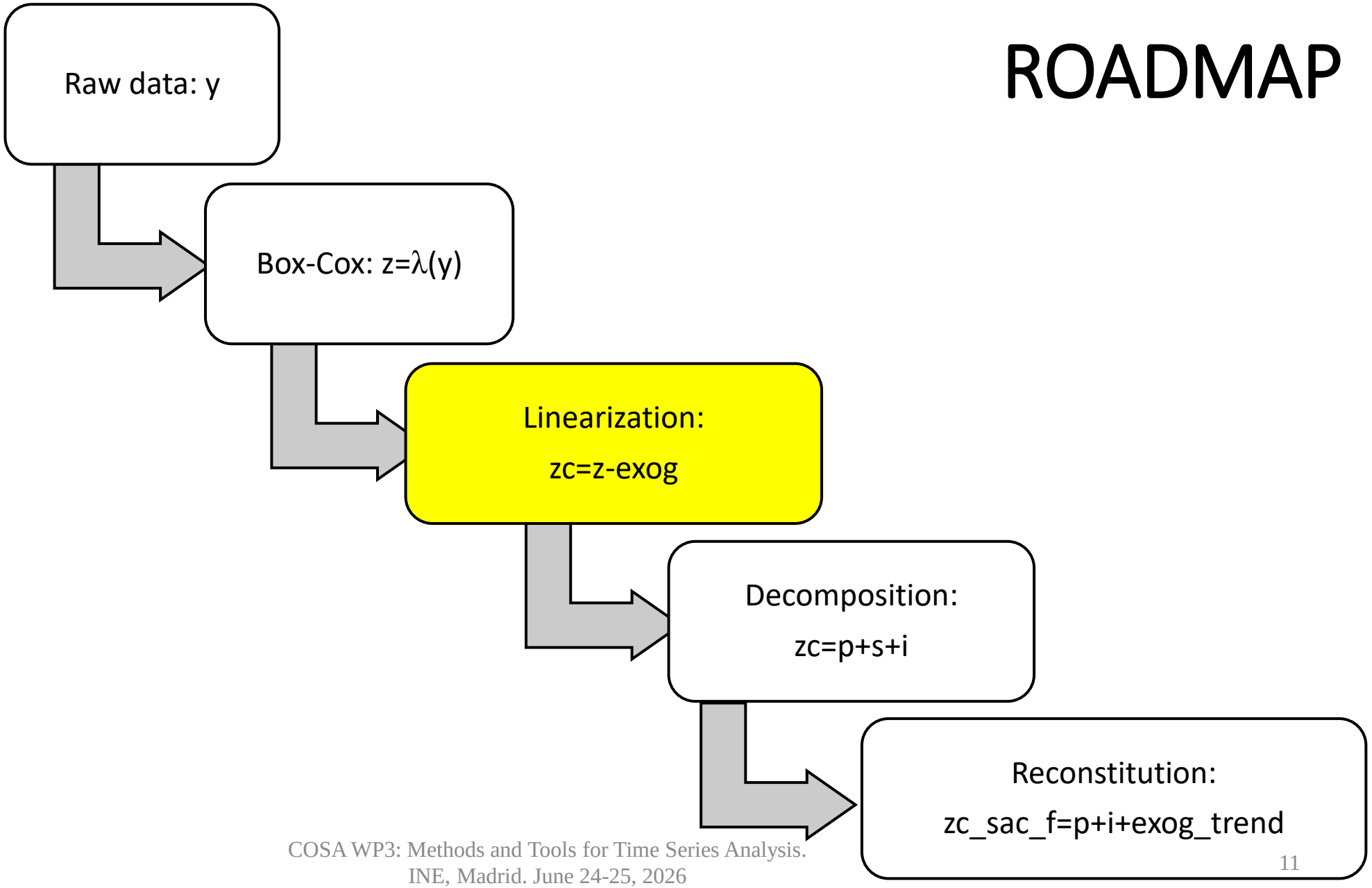

```
> print(c(res_rm$t_ratio, res_rm$t_ratio_robust))
[1] 17.17081 11.97924
```





- A refined search using the Guerrero (1993) procedure.
- A transformation seems convenient.
- For simplicity, we have selected $\lambda=0$.
- The same results are obtained with the linearized time series, a more tilted towards $\lambda=0$.

ROADMAP


Raw data: y
Box-Cox: z=λ(y)
Linearization:
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Decomposition:
zc=p+s+i
Reconstitution:
zc_sac_f=p+i+exog_trend

SUBSERIES FACTOR ANALYSIS

FACTOR ANALYSIS OF WEEKLY SUBSERIES

- We will consider a tabular form for the daily time series:

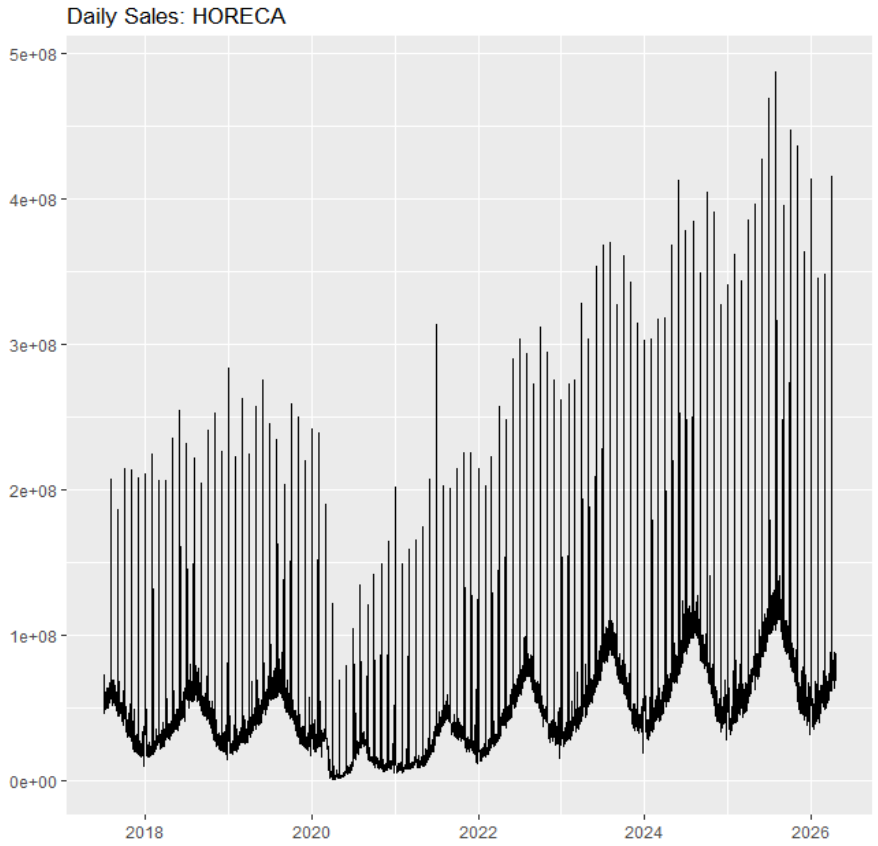
  - [Weeks] x [Day of the Week (Monday to Friday)]: Nw x 7
  - The static factor model represents each weekly subseries as the combination of a cross-section profile (the stylized weekly cycle), a common weekly factor (modulated by DoW loadings) plus a residual idiosyncratic term:

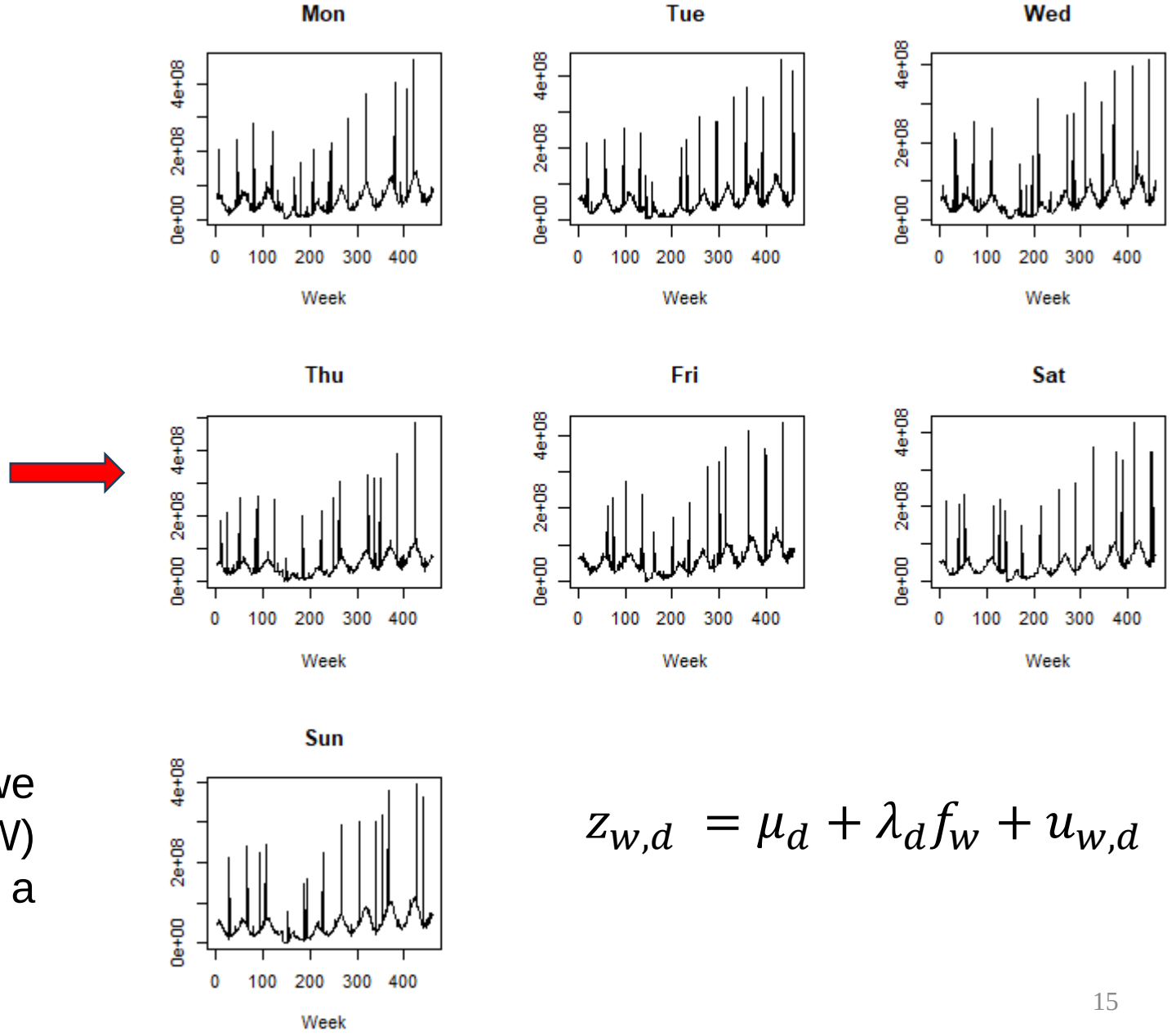
$$z_{w,d} = \mu_d + \lambda_d f_w + u_{w,d}$$

FACTOR ANALYSIS OF WEEKLY SUBSERIES

- We can extract meaningful information easily:

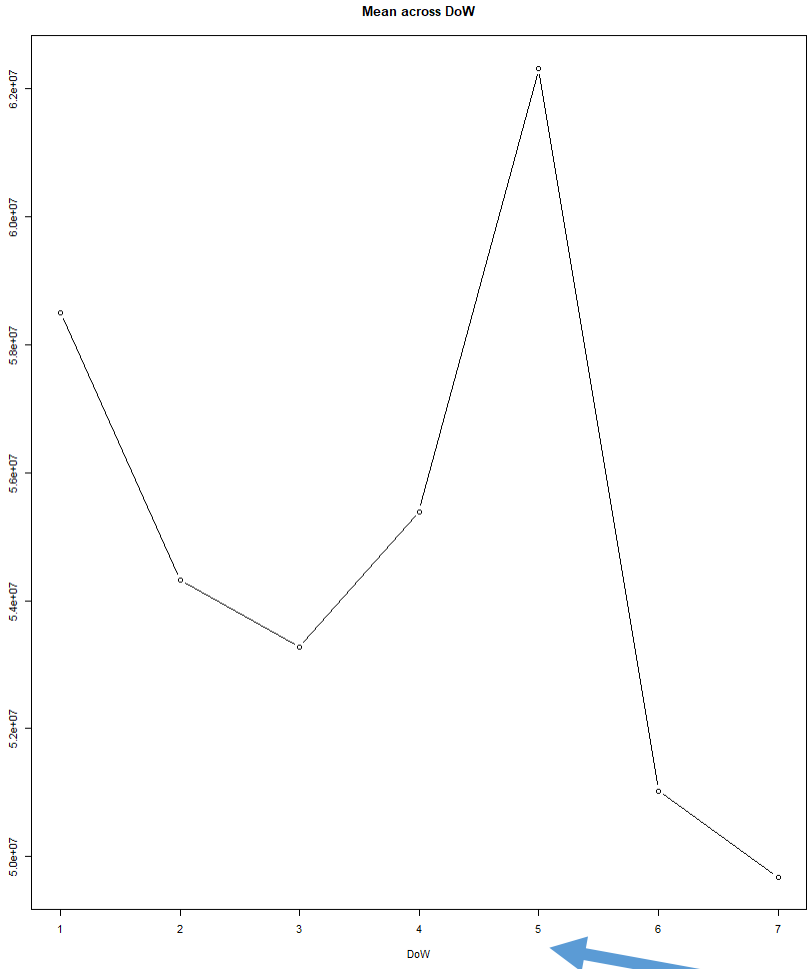
  - If the intercept (μ) is not flat → possible weekly seasonal pattern.
  - The common factor contains information about seasonal components (linked to lower frequencies) and the trend.
  - The better the fit of the 1f model, the more reliable the conclusions are.
  - The factor provides a data-driven and more informative alternative to temporal aggregation.

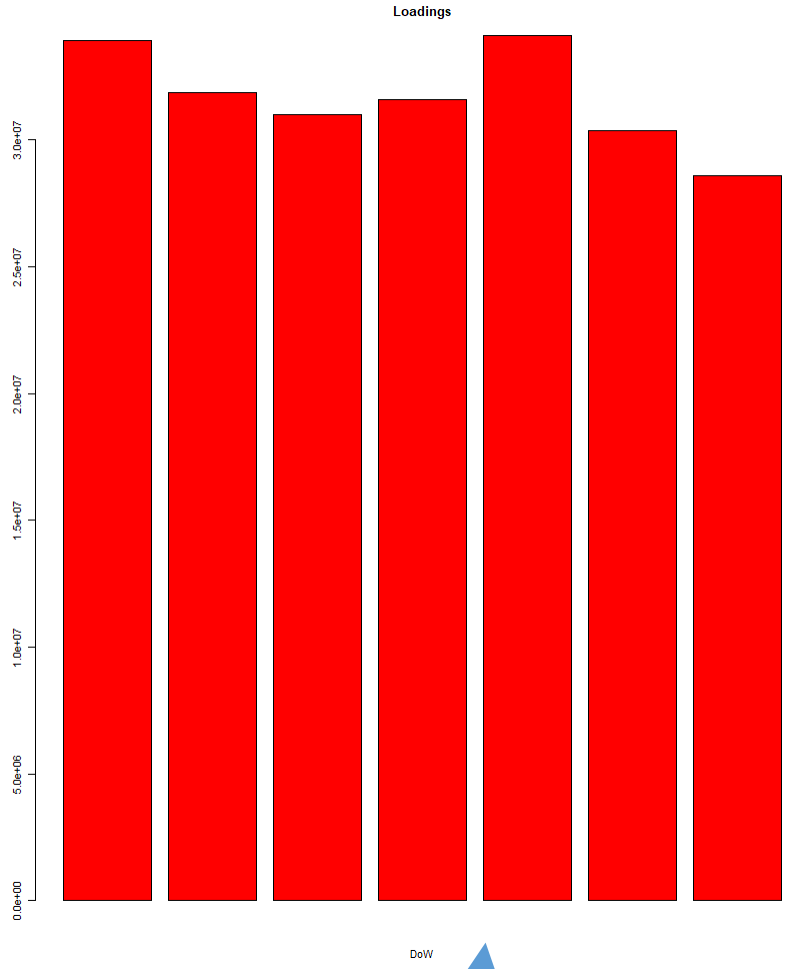


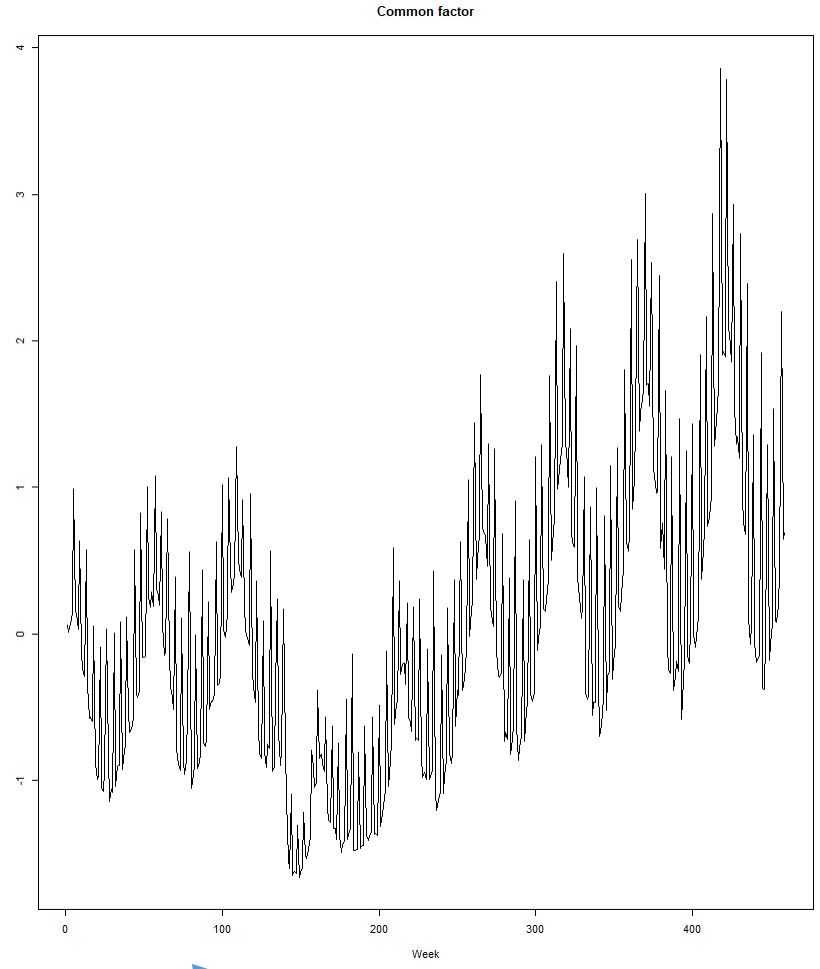


- A simple exploratory device: can we identify a day-of-the-week (DoW) profile once we have controlled for a weekly common factor?

$$z_{w,d} = \mu_d + \lambda_d f_w + u_{w,d}$$

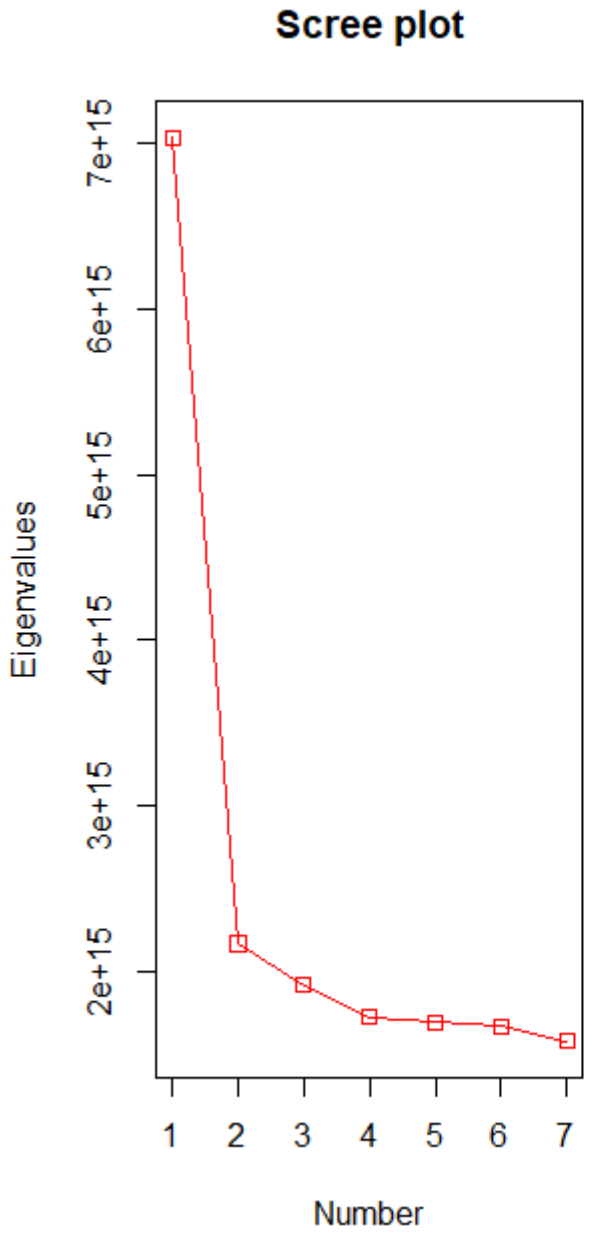


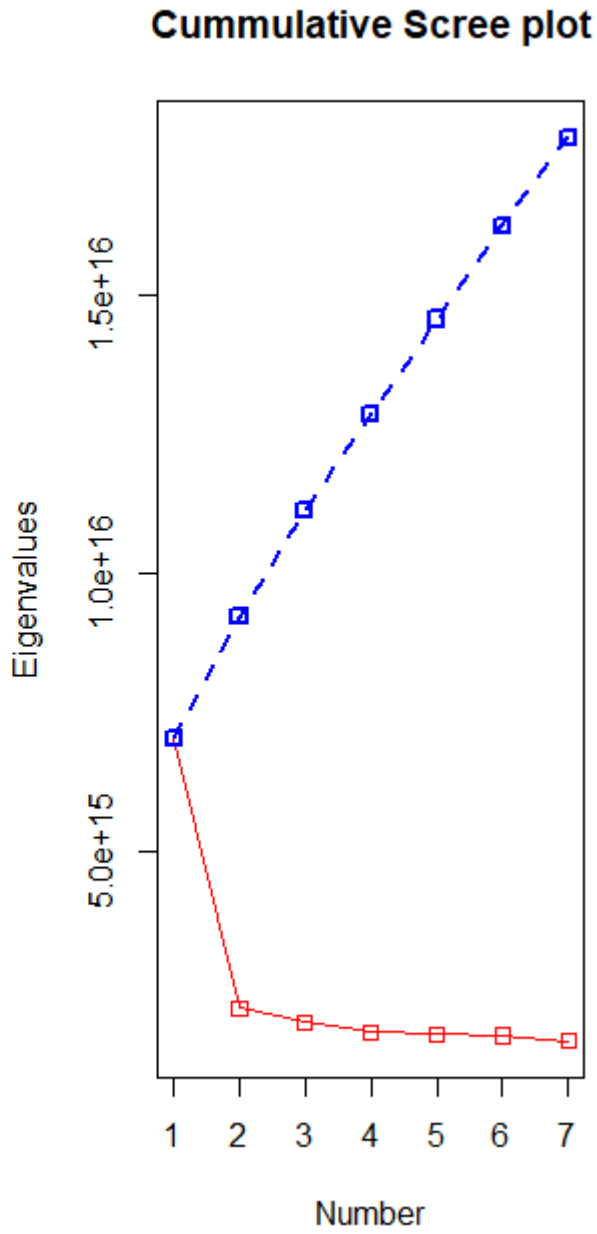




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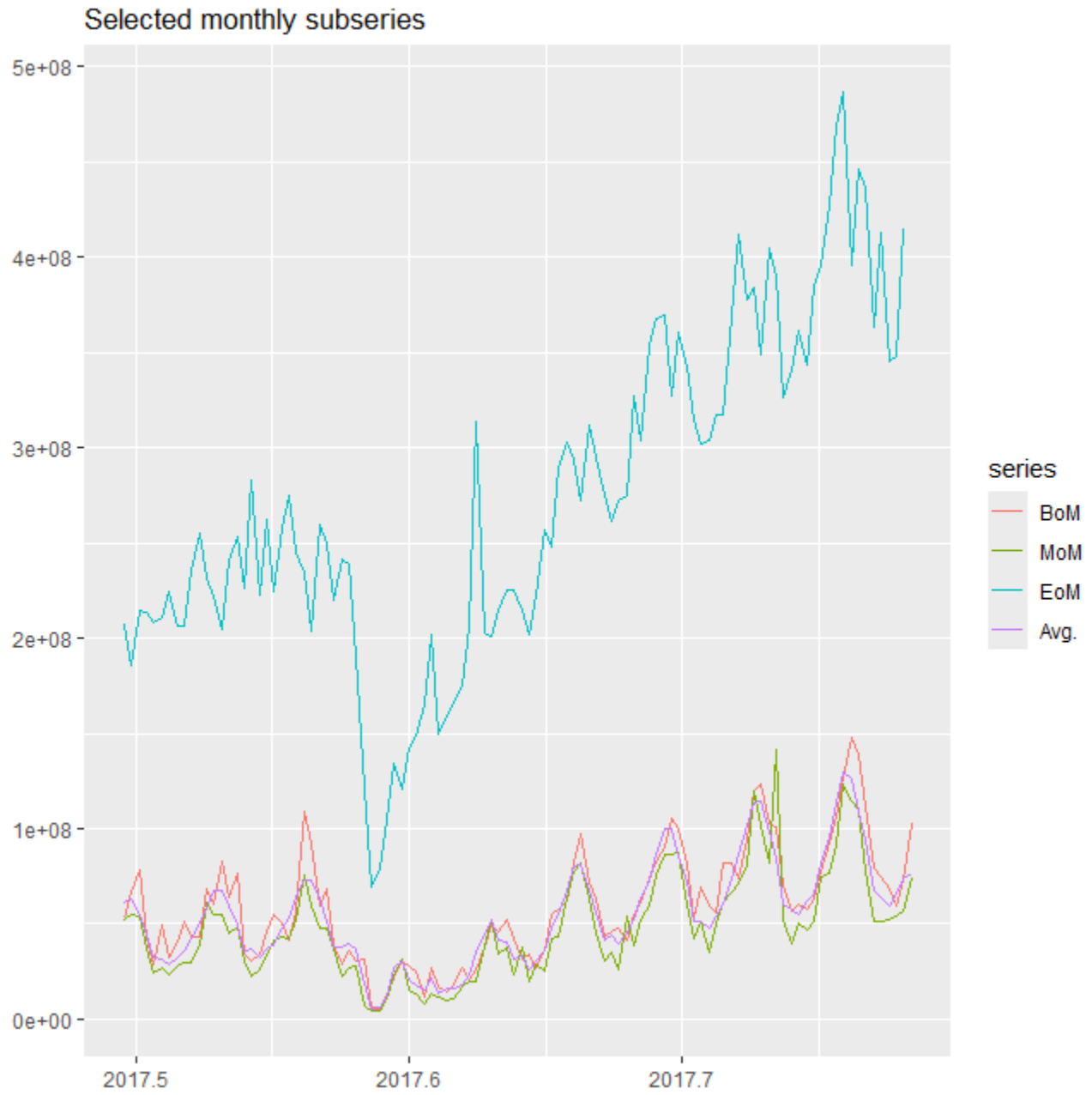


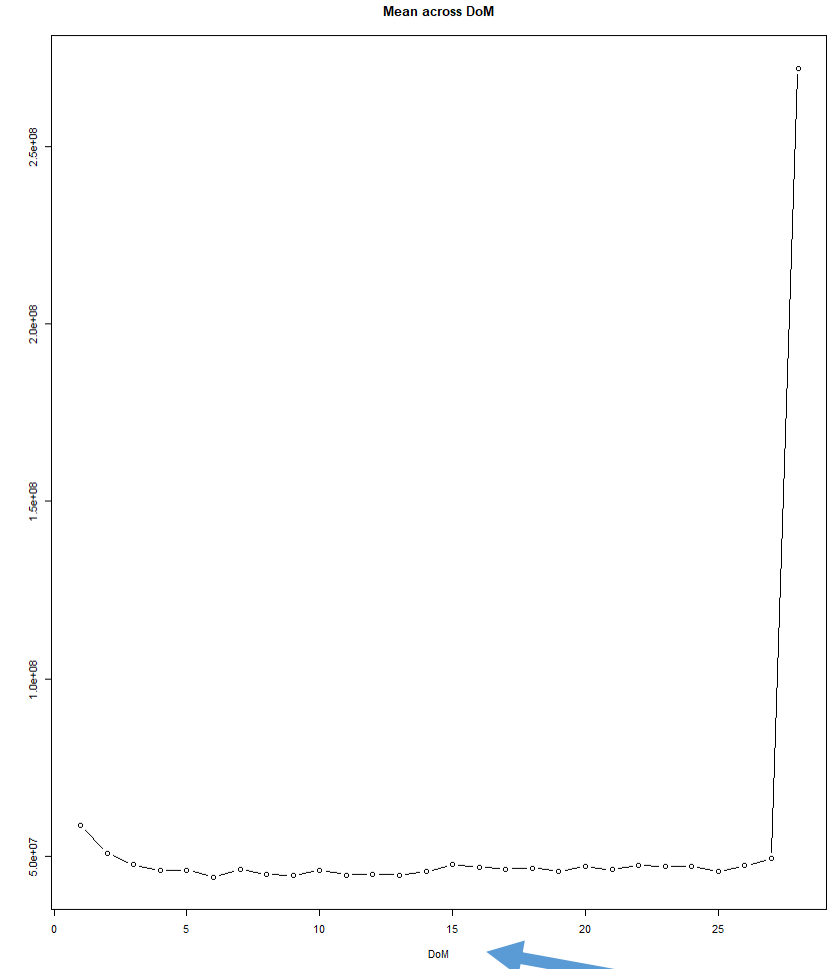
FACTOR ANALYSIS OF MONTHLY SUBSERIES

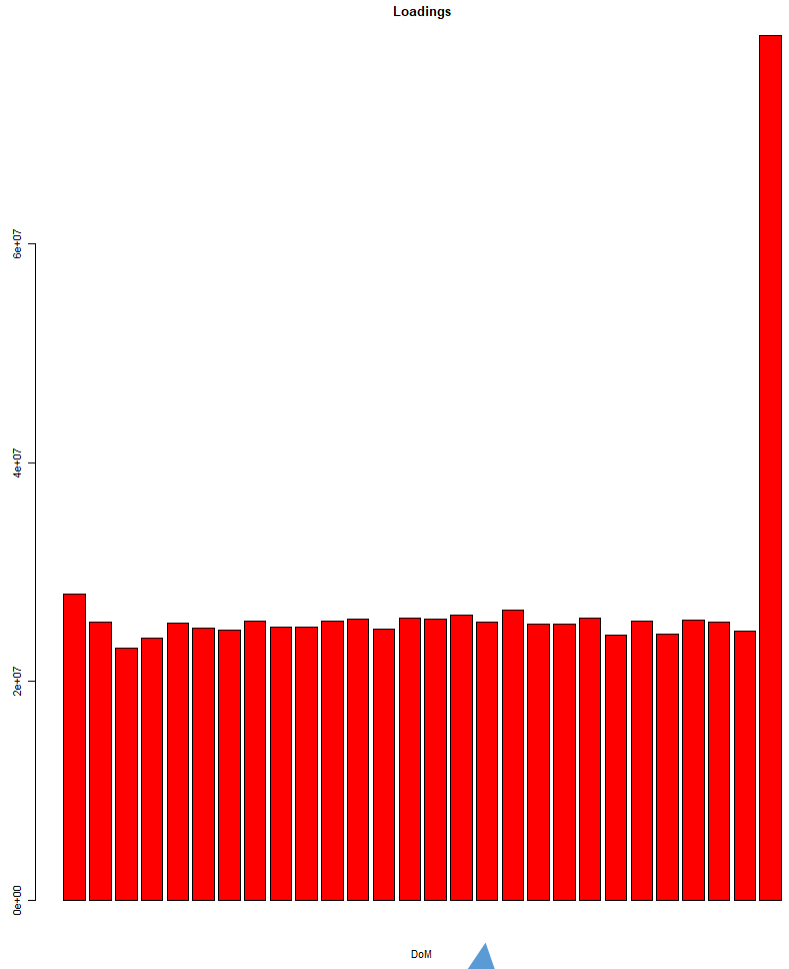
- We will consider an alternative tabular form for the daily time series:
  - [Months] x [Day of the Month (1st to 27th + EoM=End of Month)]: Nm x 28
  - The analysis is less complete than in the case of the weekly subseries due to the loss of some days (notably, those of the days 28th to 29th).
  - Again, the factor model represents each monthly subseries as the combination of cross-section profile (the simplified monthly cycle), a common monthly factor (modulated by DoM loadings) plus a residual idiosyncratic term:

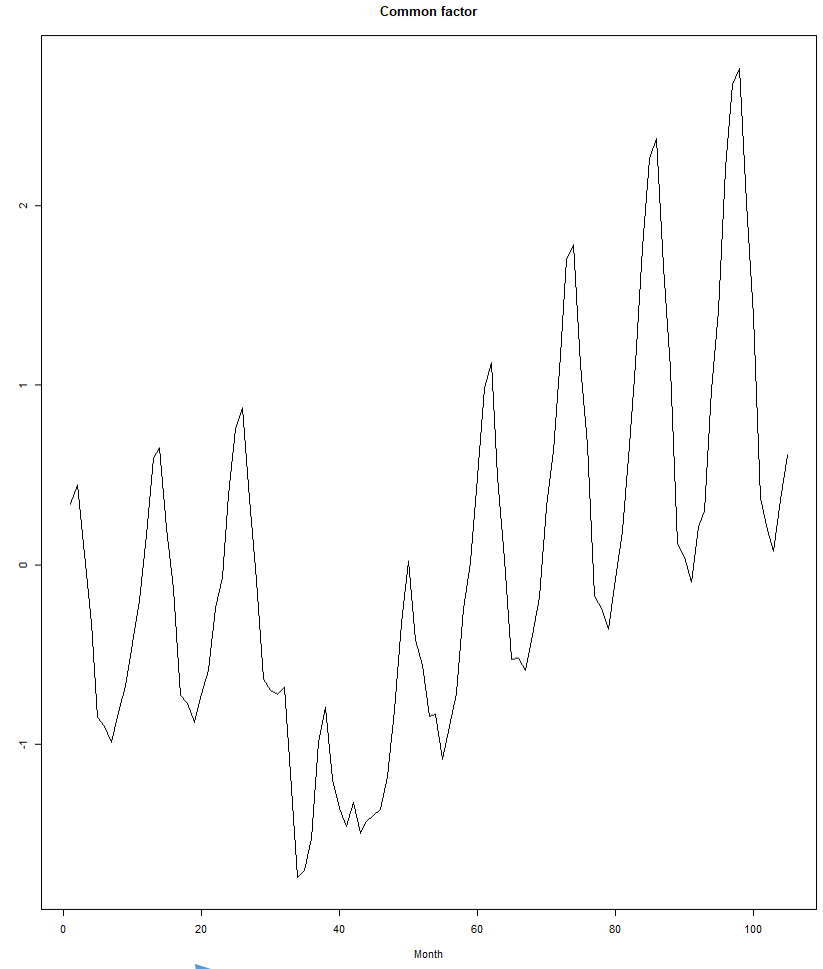
$$z_{m,d} = \mu_d + \lambda_d f_m + u_{m,d}$$

  - Its interpretation is the same as in the weekly case.


Selected monthly subseries
5e+08
4e+08
3e+08
2e+08
1e+08
0e+00
2017.5
2017.6
2017.7
series
BoM
MoM
EoM
Avg.

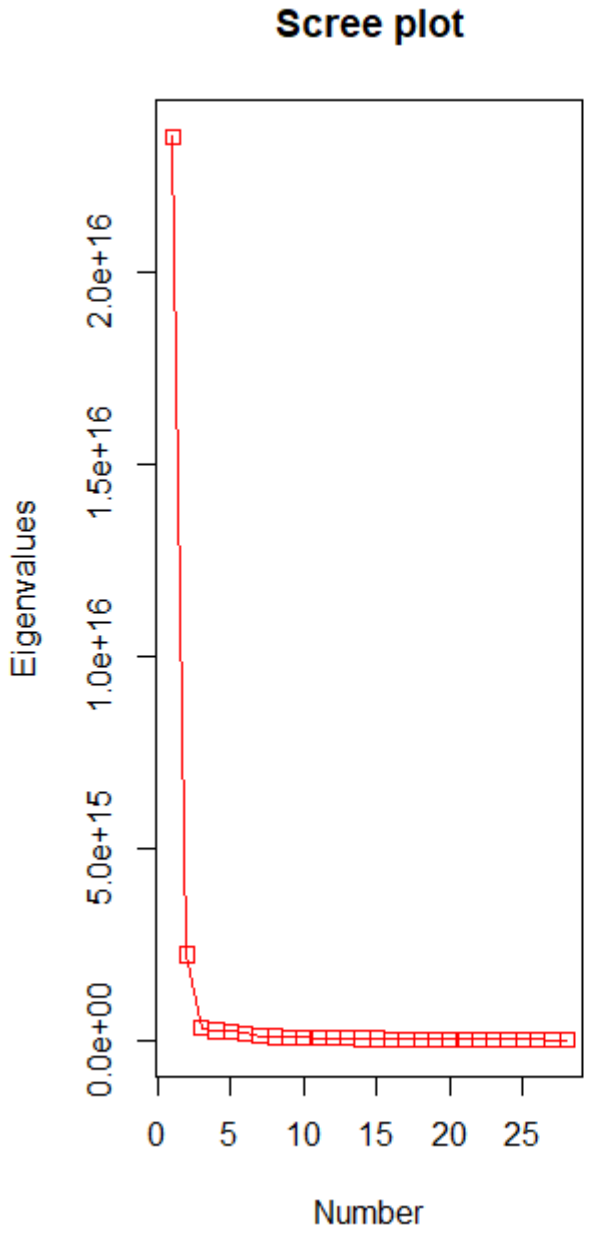


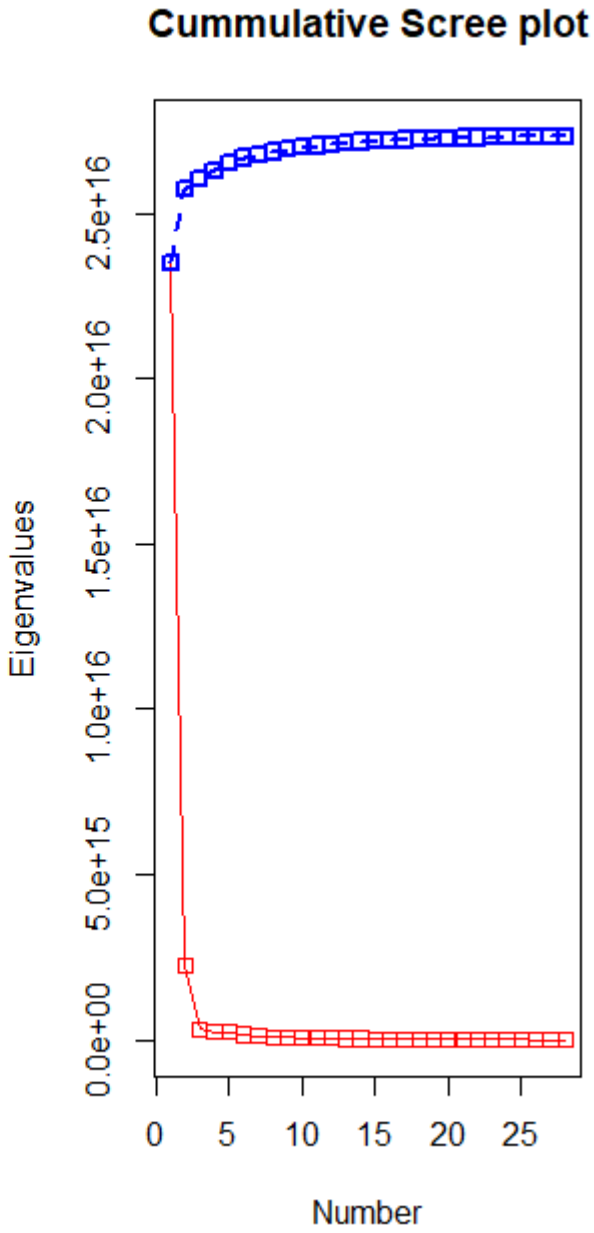




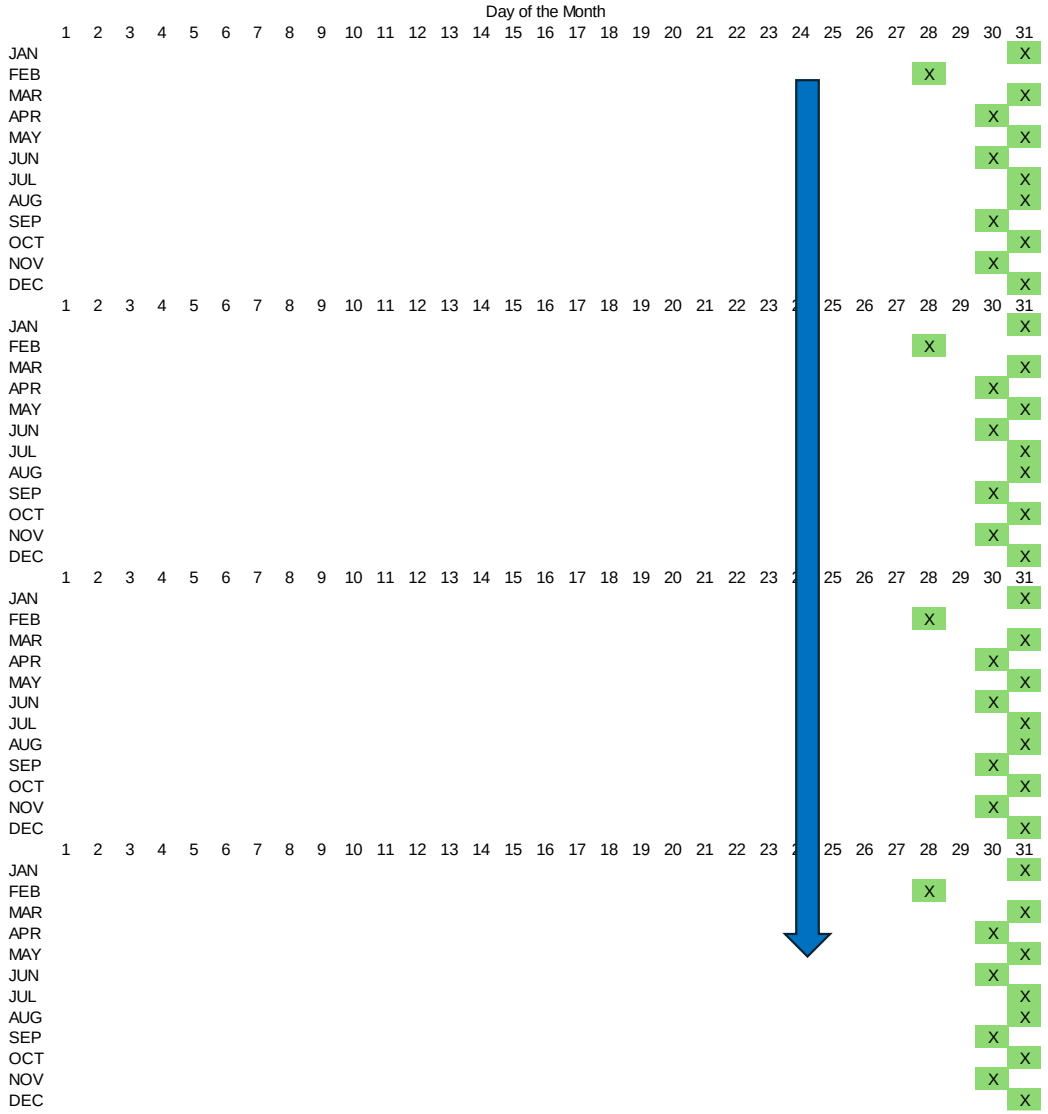
$$z_{m,d} = \mu_d + \lambda_d f_m + u_{m,d}$$

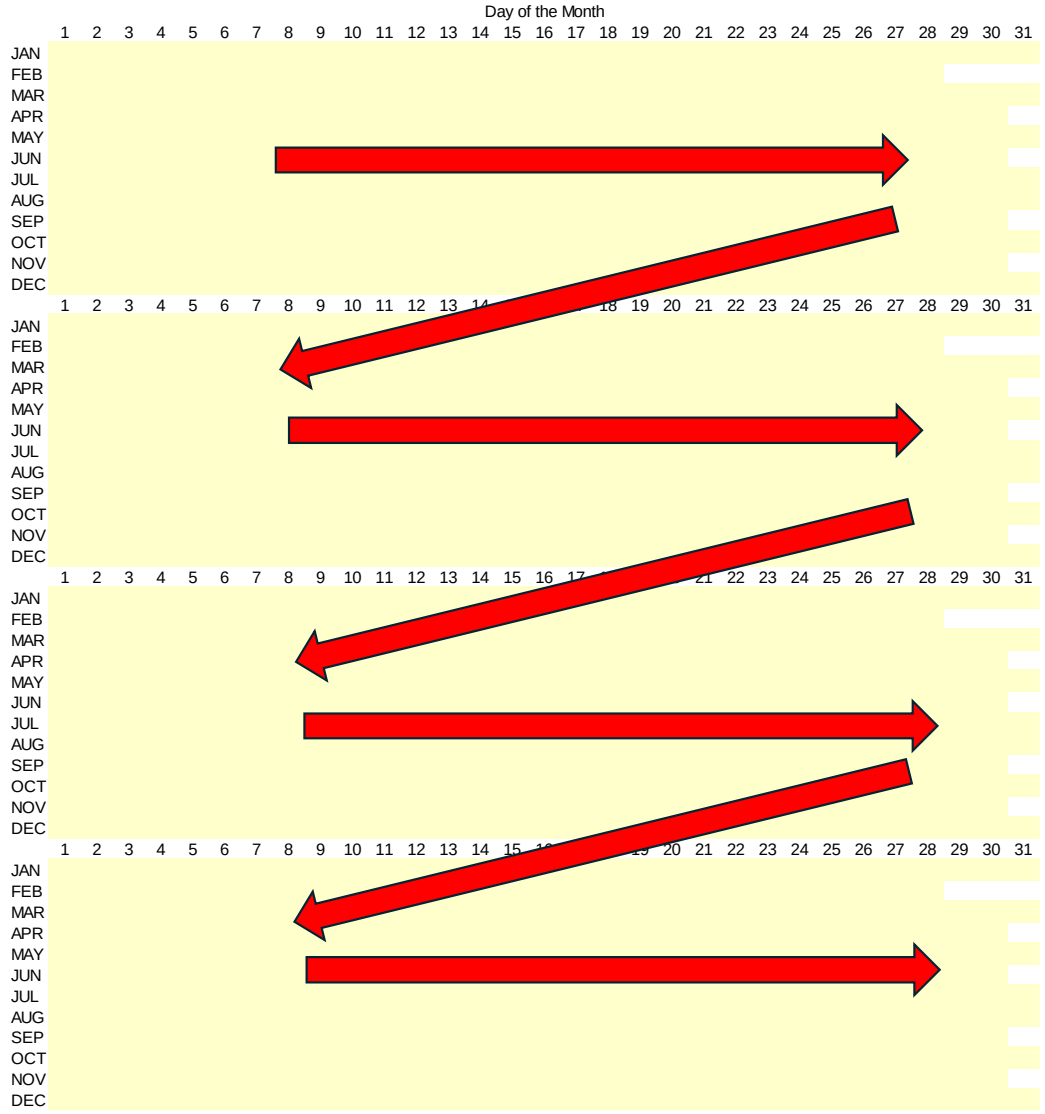
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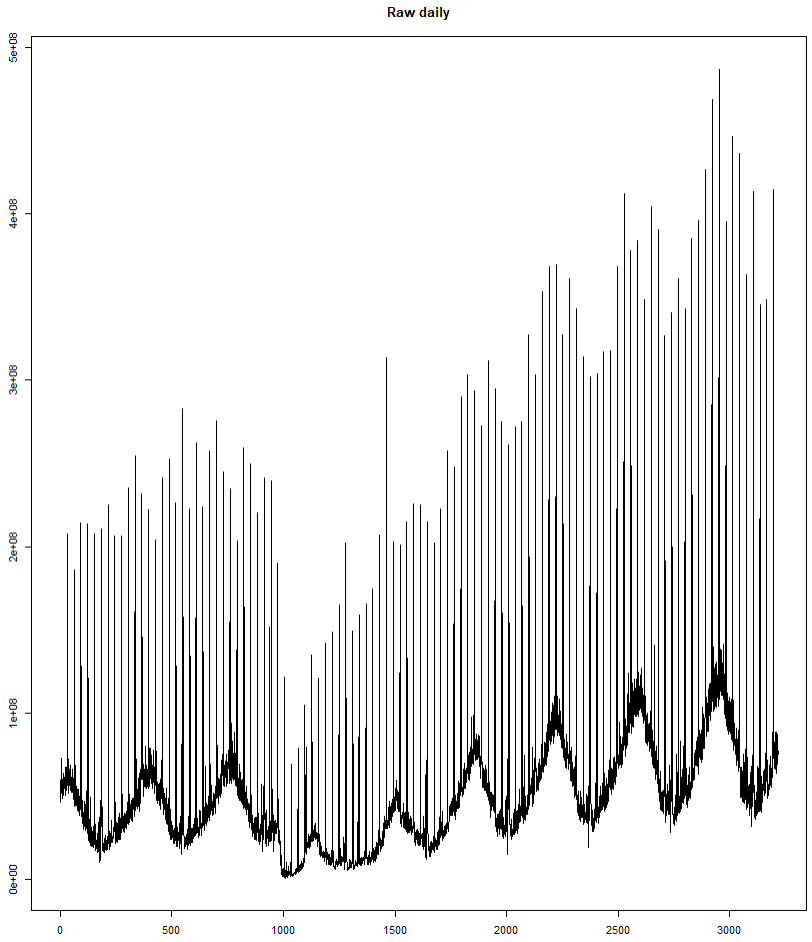


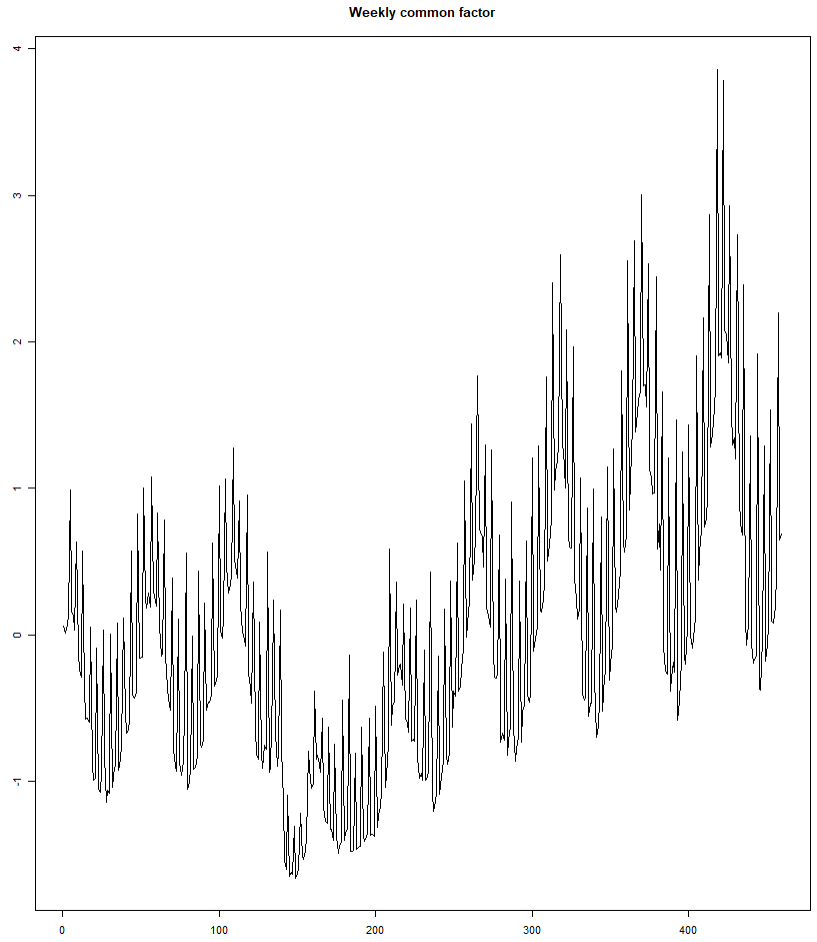


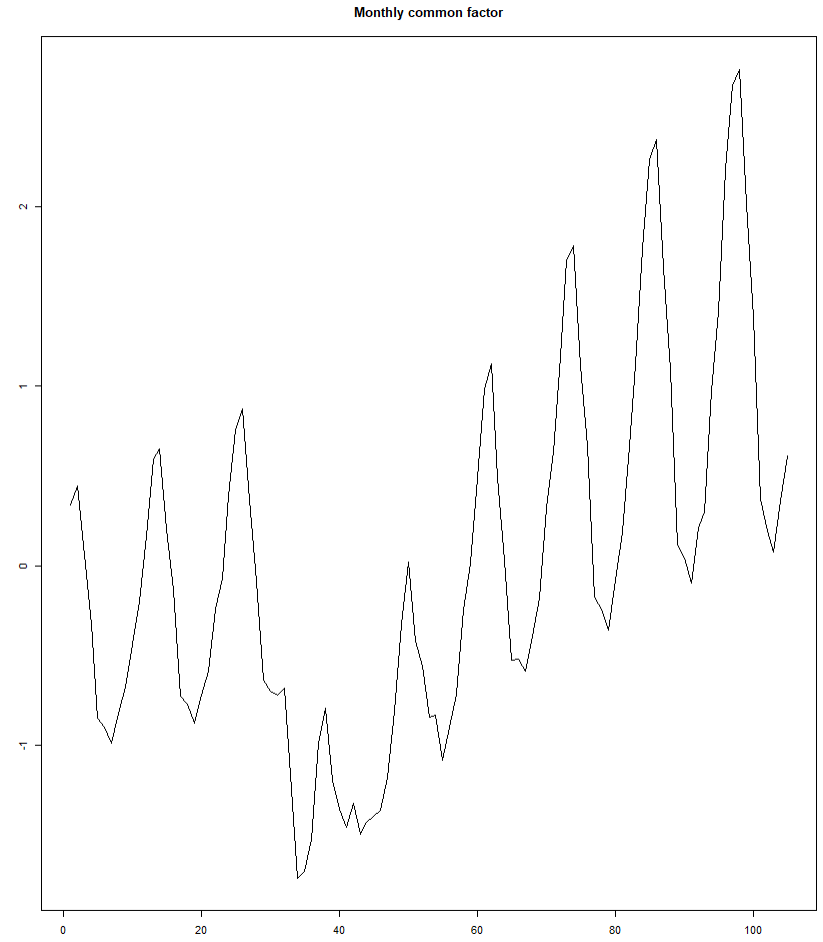
$$z_d = z_daily_d + sel_d \, Z_monthly_T \; ??$$






Raw daily
0e+00
1e+08
2e+08
3e+08
4e+08
5e+08
0
500
1000
1500
2000
2500
3000


Weekly common factor
-1
0
1
2
3
4
0
100
200
300
400


Monthly common factor
-1
0
1
2
0
20
40
60
80
100

	Does it have…			
	Seasonality: weekly	Seasonality: monthly	Seasonality: annual	Trend
Common factor of monthly subseries	NO	NO	**YES**	**YES**
Common factor of weekly subseries	NO	**YES**	**YES**	**YES**
Raw daily time series	**YES**	**YES**	**YES**	**YES**

→ Rough estimate of the (mean) weekly (monthly) cycle.

MONTHLY SEASONALITY

- A possibility is to include exogenous dummies to take into account this pattern and pre-process it.
- Hyndman (2017):

The best approach is to consider what is causing the monthly seasonality, and try to allow for it explicitly using regressor variables. If there is an end-of-month effect, then a dummy variable indicating which day(s) are at the end of the month would effectively handle it. If there is a regular event on a specific day in the month (such as the second Tuesday), then again a dummy variable would be best used.

MONTHLY SEASONALITY

- We will consider the monthly pattern by means of six exogenous dummy regressors:

  - BoM (Begin of the Month = 1st day)
  - MoM (~Middle of the Month = 15th day)
  - EoM (End of the Month = {28, 29, 30 or 31},
  - Interaction of XoM with the weekends.

CALENDAR EFFECTS

- In addition, we will pre-process the data using a collection of exogenous regressors designed to control for the effects of national holidays, “big” regional holidays (Andalucía, Cataluña and Madrid, the largest Spanish regions) and Easter.

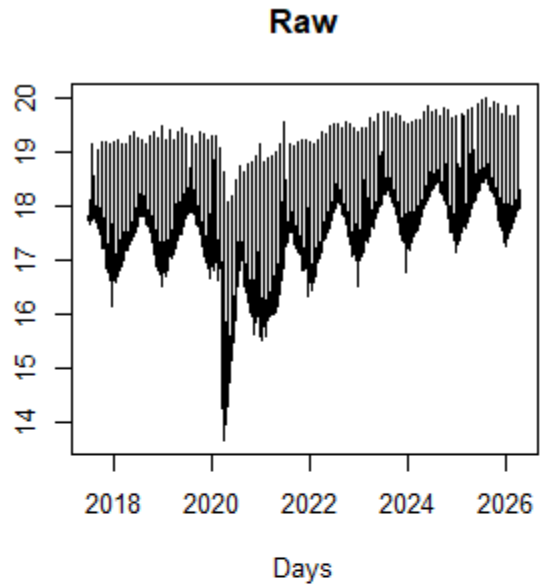
CALENDAR EFFECTS

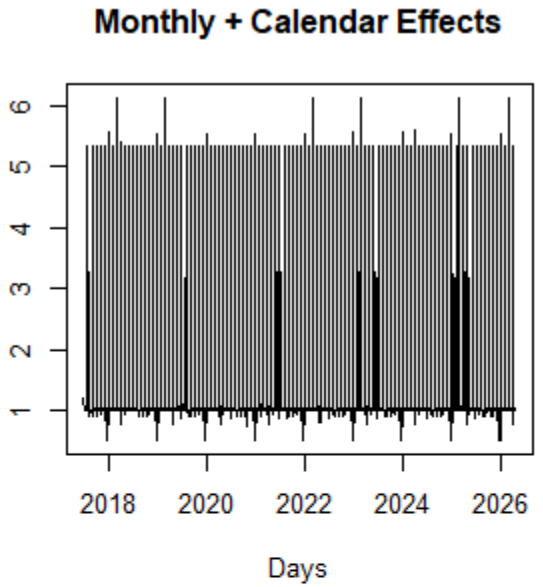
- National holidays according to the official labor calendar, national and regional (Official State Gazette, October previous year).
- Easter: from Good Thursday to Easter Monday (5 regressors).
- Regional holidays according to the Official State Gazette: Andalucía (1), Cataluña (3) and Madrid (2).

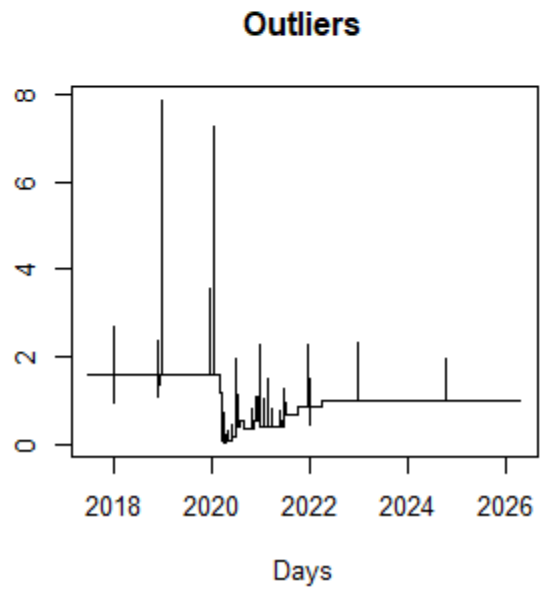
rjd3hfreq CODE

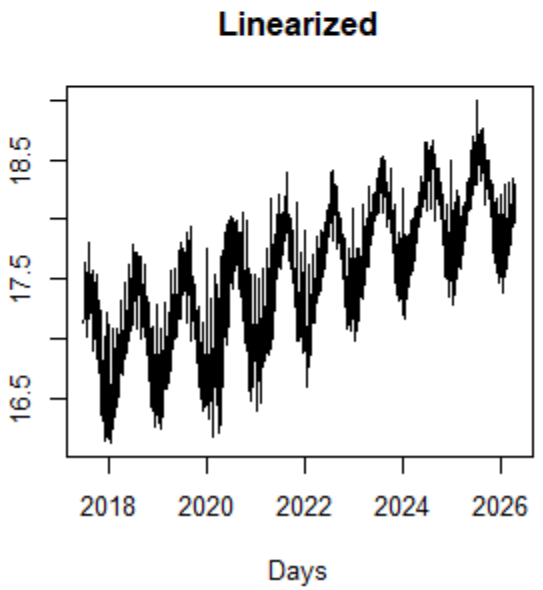
```
res_AMB<- rjd3highfreq::fractionalAirlineEstimation(
  z_ts,
  x = X_ts[,-(29:31)],                          # X regressors without CV19 a priori dummies
  periods = c(7, 365.2425),
  ndiff = 2,
  ar = FALSE,
  # outliers = NULL,
  outliers = c("ao","wo","ls"),    # type of outliers detected
  criticalValue = 0,               # automatically set
  precision = 1e-12,
  approximateHessian = TRUE,
  nfcasts = 0,
  log = TRUE,
  y_time = NULL
)
```

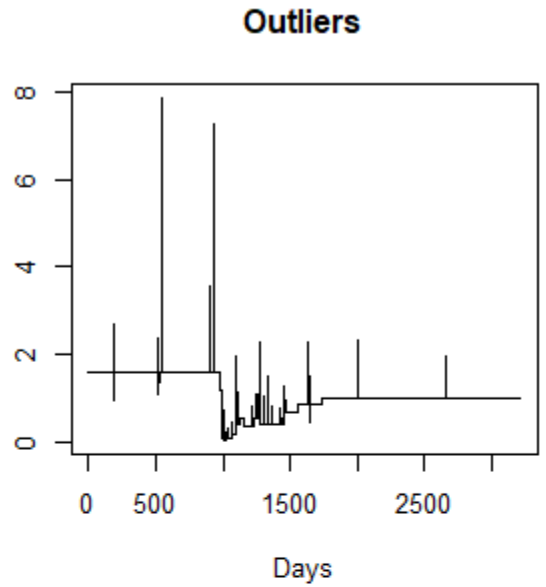
		AMB	
	Variable	**beta**	**t_ratio**
01-ene	new year	**-0,24**	**-4,45**
06-ene	kings	**-0,20**	**-4,46**
01-may	labor day	**-0,14**	**-2,82**
15-ago	assumption	**-0,13**	**-2,78**
12-oct	national fest	-0,07	-1,57
01-nov	all saints	**-0,28**	**-6,02**
06-dic	constitution	**-0,17**	**-3,75**
08-dic	immaculate	**-0,18**	**-4,07**
25-dic	christmas	**-0,67**	**-10,49**
24-dic	christmas eve	**-0,64**	**-10,38**
31-dic	new y eve	0,04	0,72
Easter	good Thurs	0,00	0,07
Easter	good Friday	**-0,25**	**-5,73**
Easter	good Sat	0,01	0,28
Easter	Easter	0,05	1,08
Easter	good Mon	0,07	1,64
CCAA big	and	**0,14**	**3,05**
CCAA big	cat	-0,08	-1,83
CCAA big	cat1	**-0,11**	**-2,08**
CCAA big	cat2	-0,07	-1,39
CCAA big	mad	-0,05	-1,04
CCAA big	mad1	-0,05	-0,93
Monthly_seas	d1	**0,23**	**12,00**
	d15	**0,06**	**3,50**
	d_last	**1,67**	**89,47**
	d1_we	-0,05	-1,79
	d15_we	-0,02	-0,59
	d_last_we	0,00	0,03

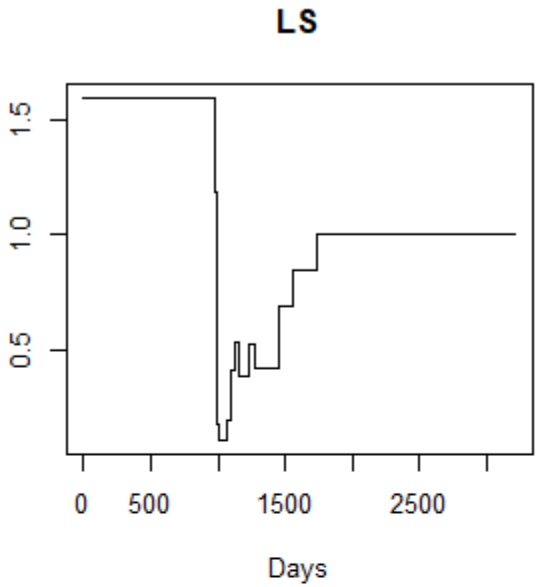

Raw
20
19
18
17
16
15
14
2018
2020
2022
2024
2026
Days

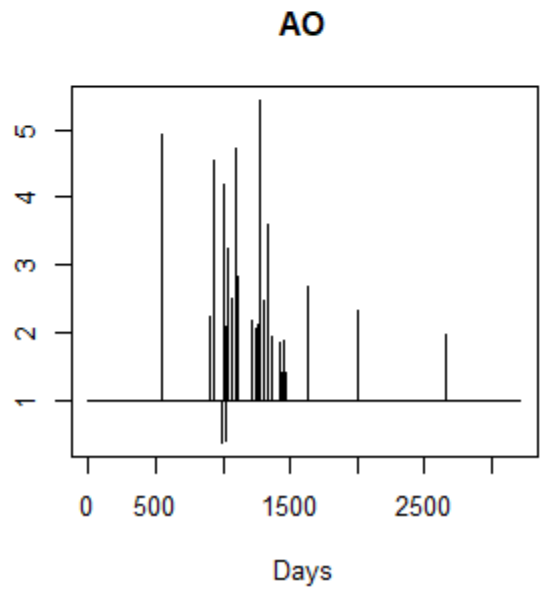

Monthly + Calendar Effects
6
5
4
3
2
1
2018
2020
2022
2024
2026
Days

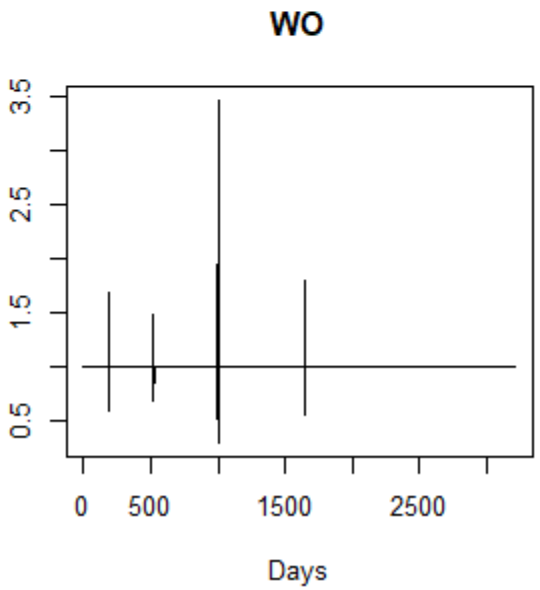

Outliers
8
6
4
2
0
2018
2020
2022
2024
2026
Days


Linearized
18.5
17.5
16.5
2018
2020
2022
2024
2026
Days

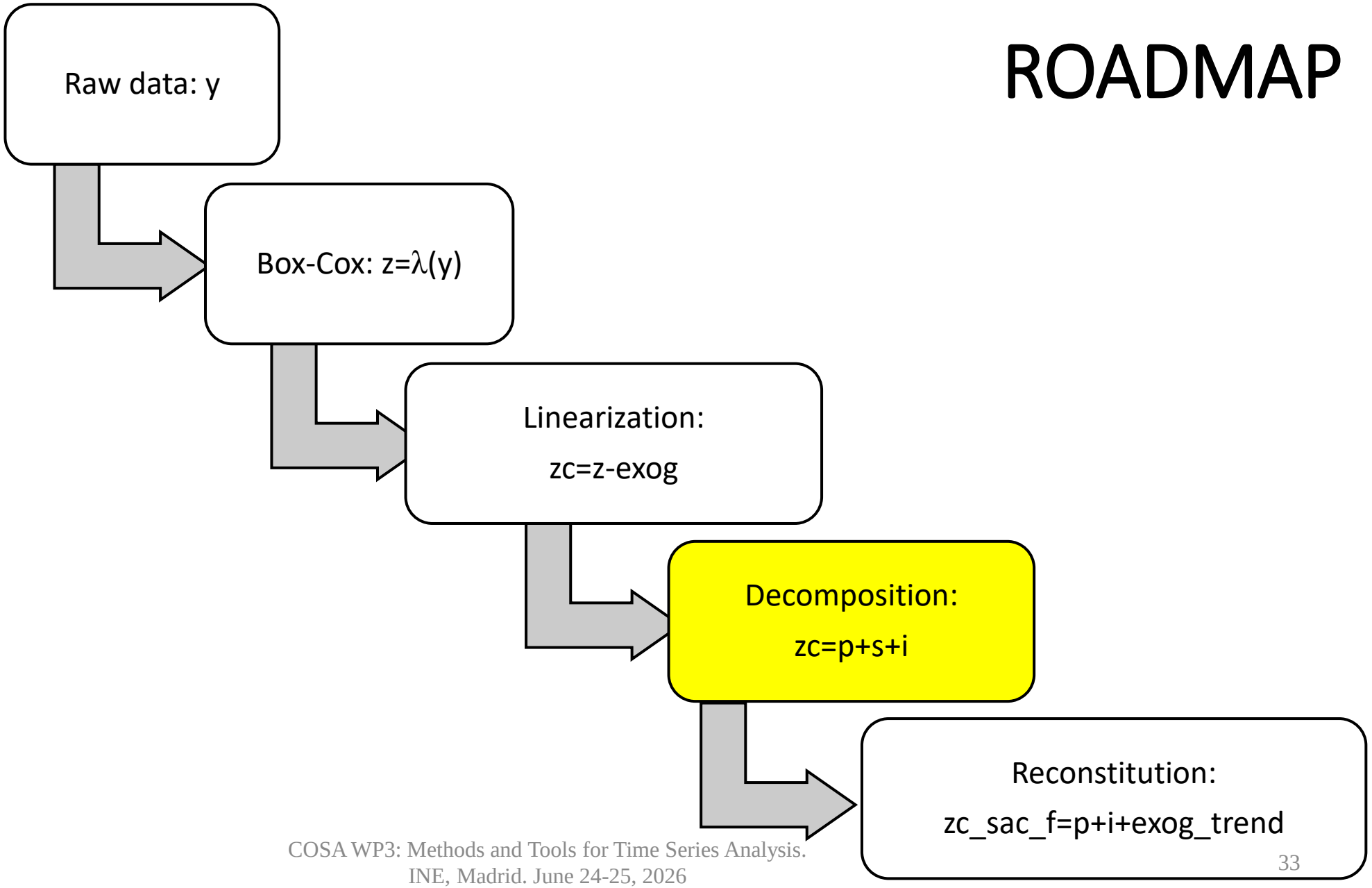

Outliers
0
2
4
6
8
0
500
1500
2500
Days


LS
0.5
1.0
1.5
0
500
1500
2500
Days


AO
1
2
3
4
5
0
500
1500
2500
Days


WO
0.5
1.5
2.5
3.5
0
500
1500
2500
Days

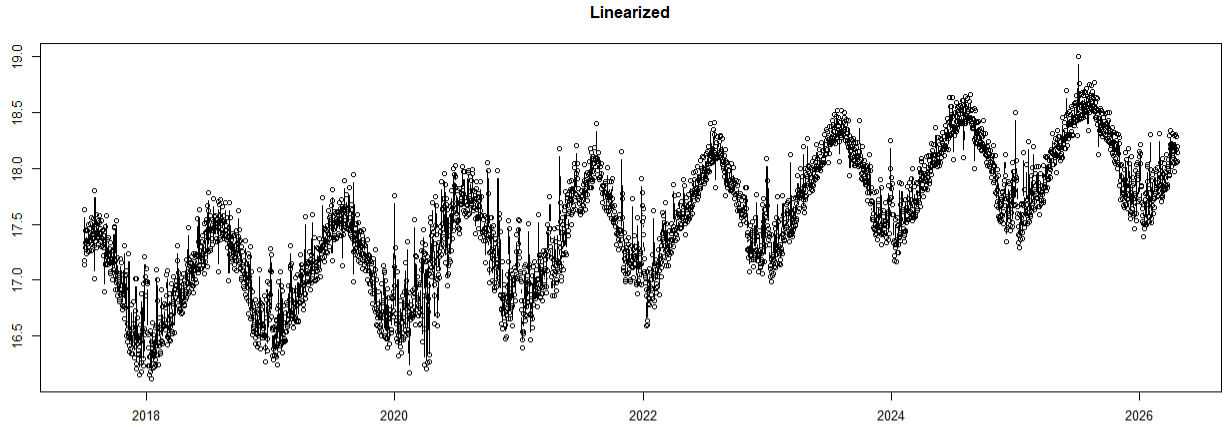
ROADMAP

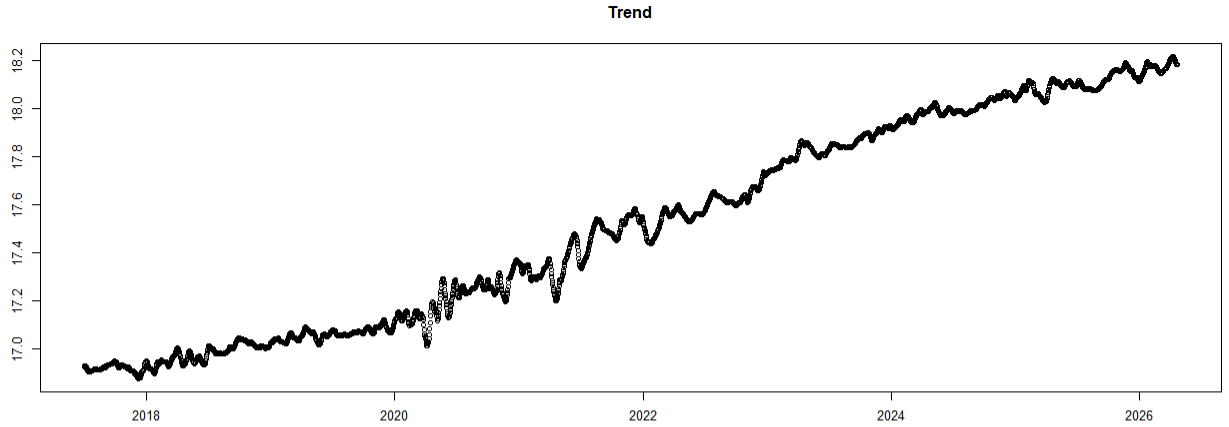

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Decomposition:
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Reconstitution:
zc_sac_f=p+i+exog_trend

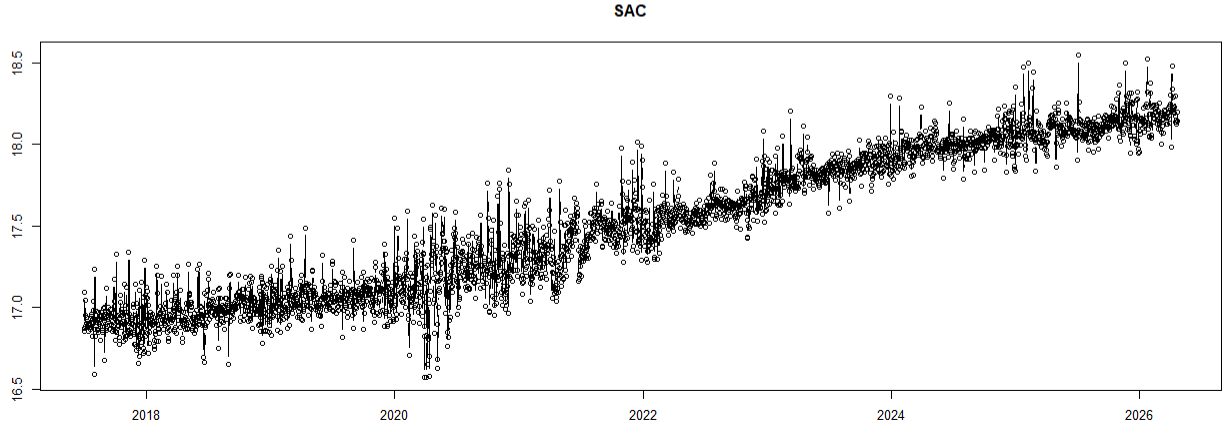
```
##############################################################
#          STEP 2: DECOMPOSITION: AMB                        #
##############################################################

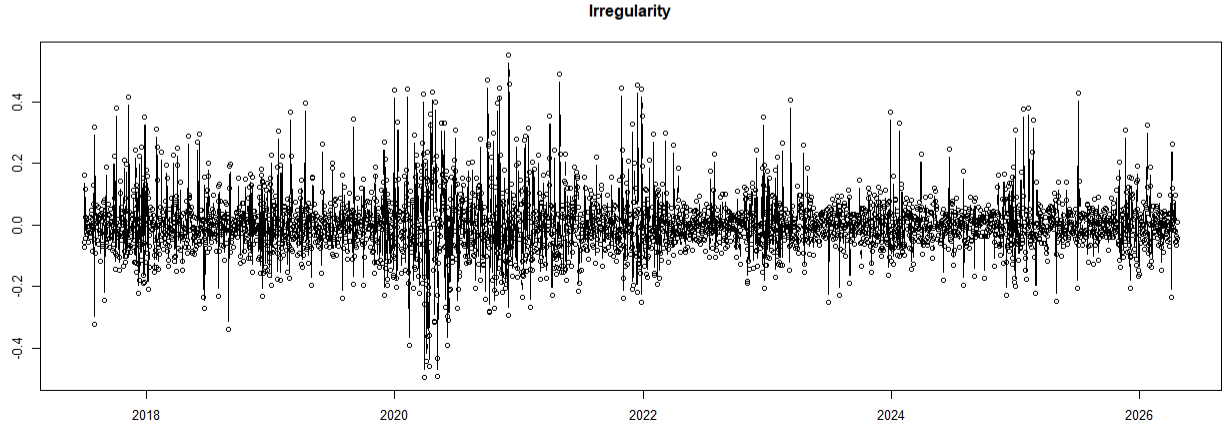
res_AMB.dow <- rjd3highfreq::multiAirlineDecomposition(
  y = res_AMB$model$linearized, # Input time series
  period = 7, # weekly frequency
  log = FALSE)

res_AMB.doy <- rjd3highfreq::multiAirlineDecomposition(
  y = res_AMB.dow$decomposition$sa, # DOW-adjusted linearised data
  period = 365.2425, # annual frequency
  log = FALSE)
```

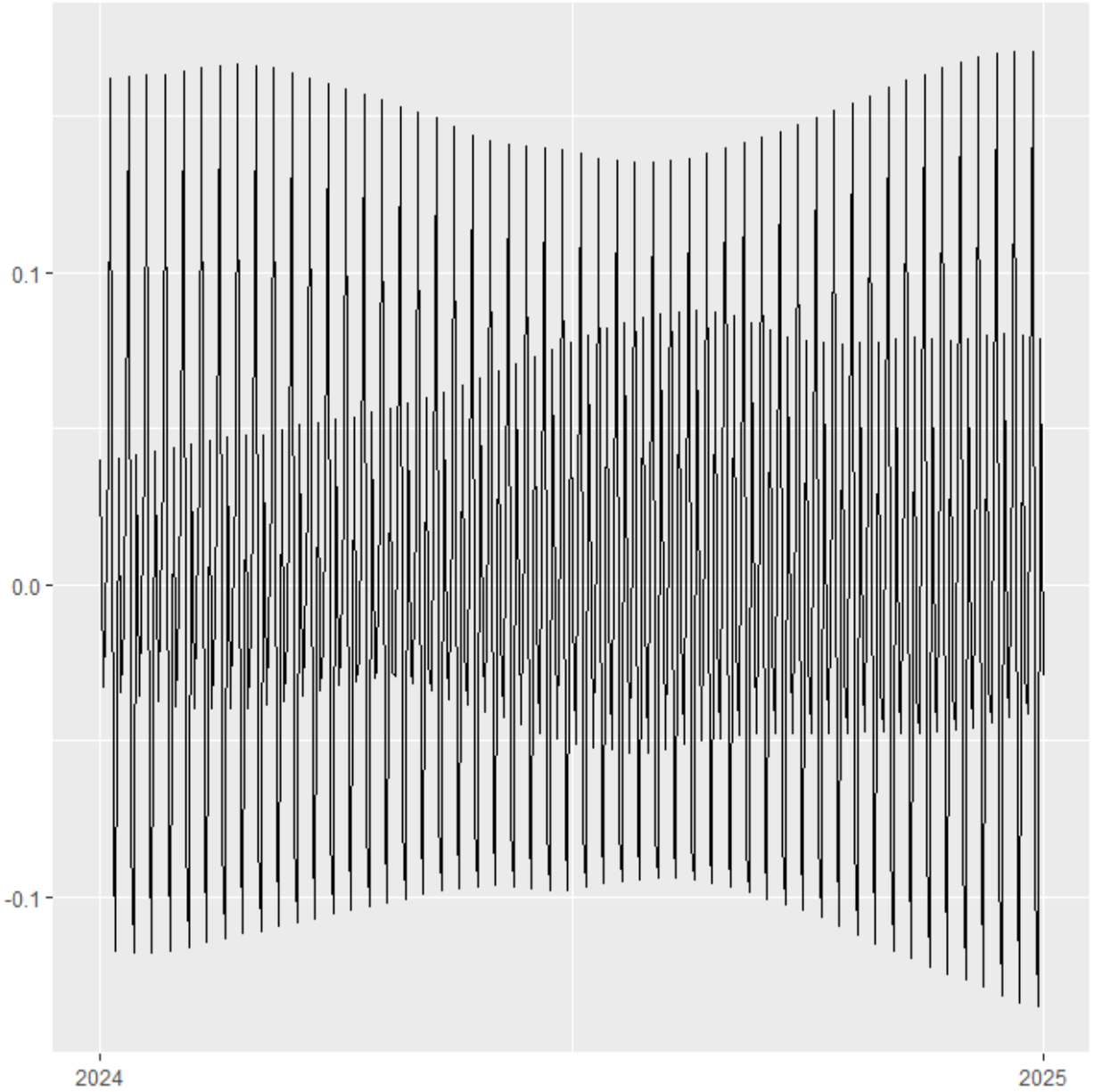

Linearized
19.0
18.5
18.0
17.5
17.0
16.5
2018
2020
2022
2024
2026


Trend
18.2
18.0
17.8
17.6
17.4
17.2
17.0
2018
2020
2022
2024
2026

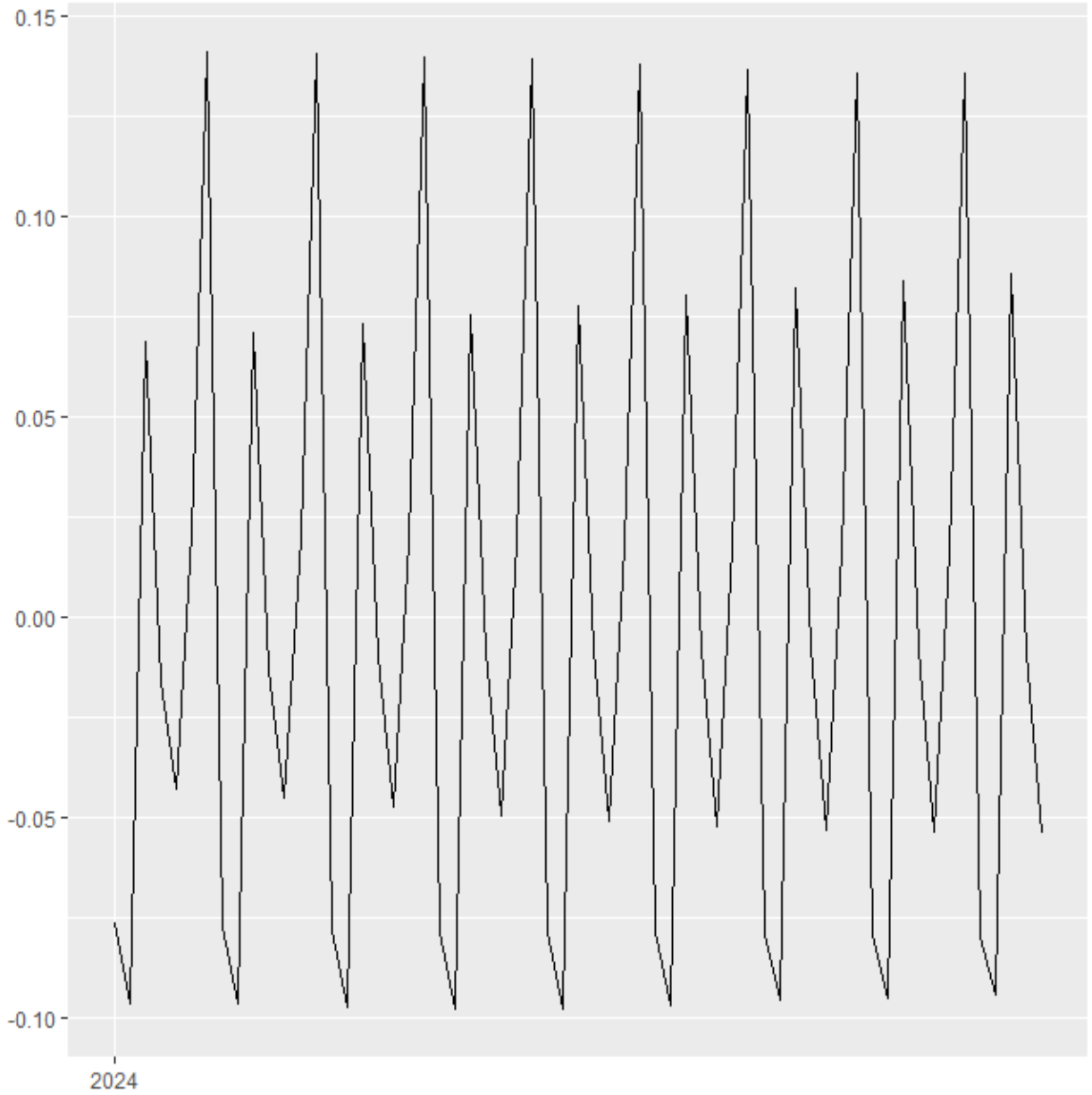

SAC
18.5
18.0
17.5
17.0
16.5
2018
2020
2022
2024
2026

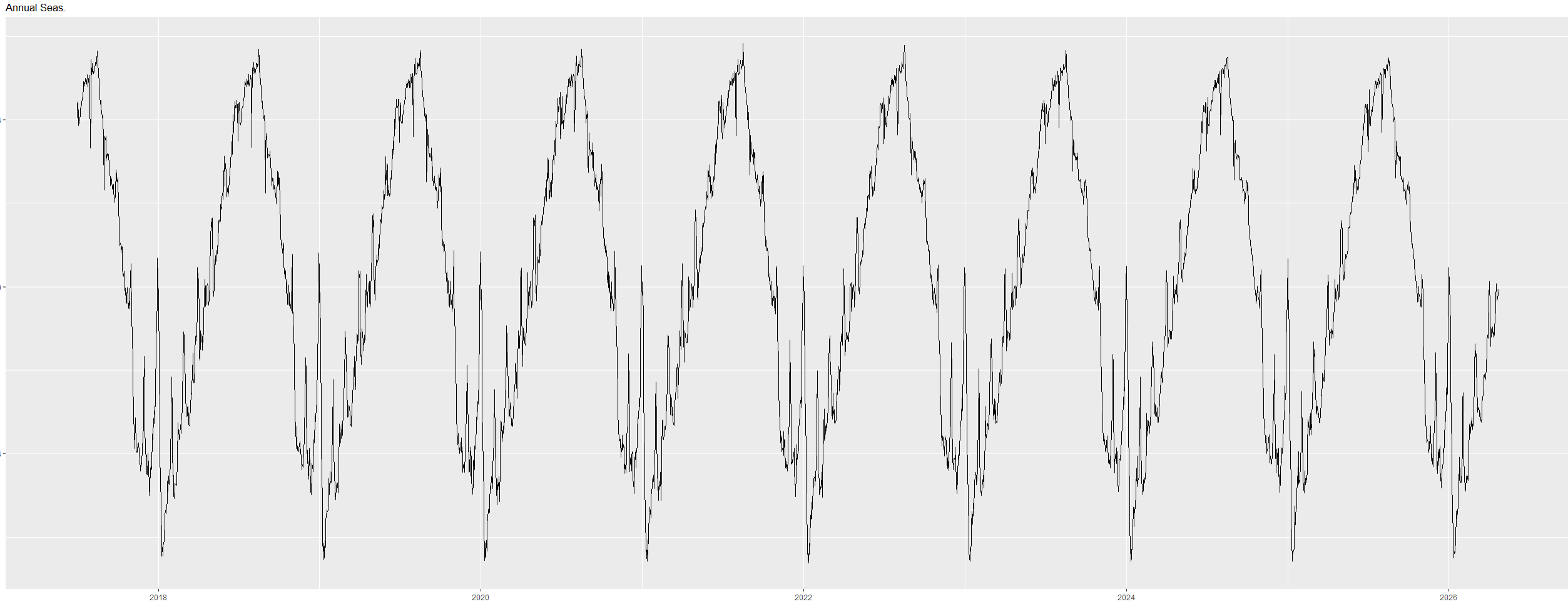

Irregularity
0.4
0.2
0.0
-0.2
-0.4
2018
2020
2022
2024
2026

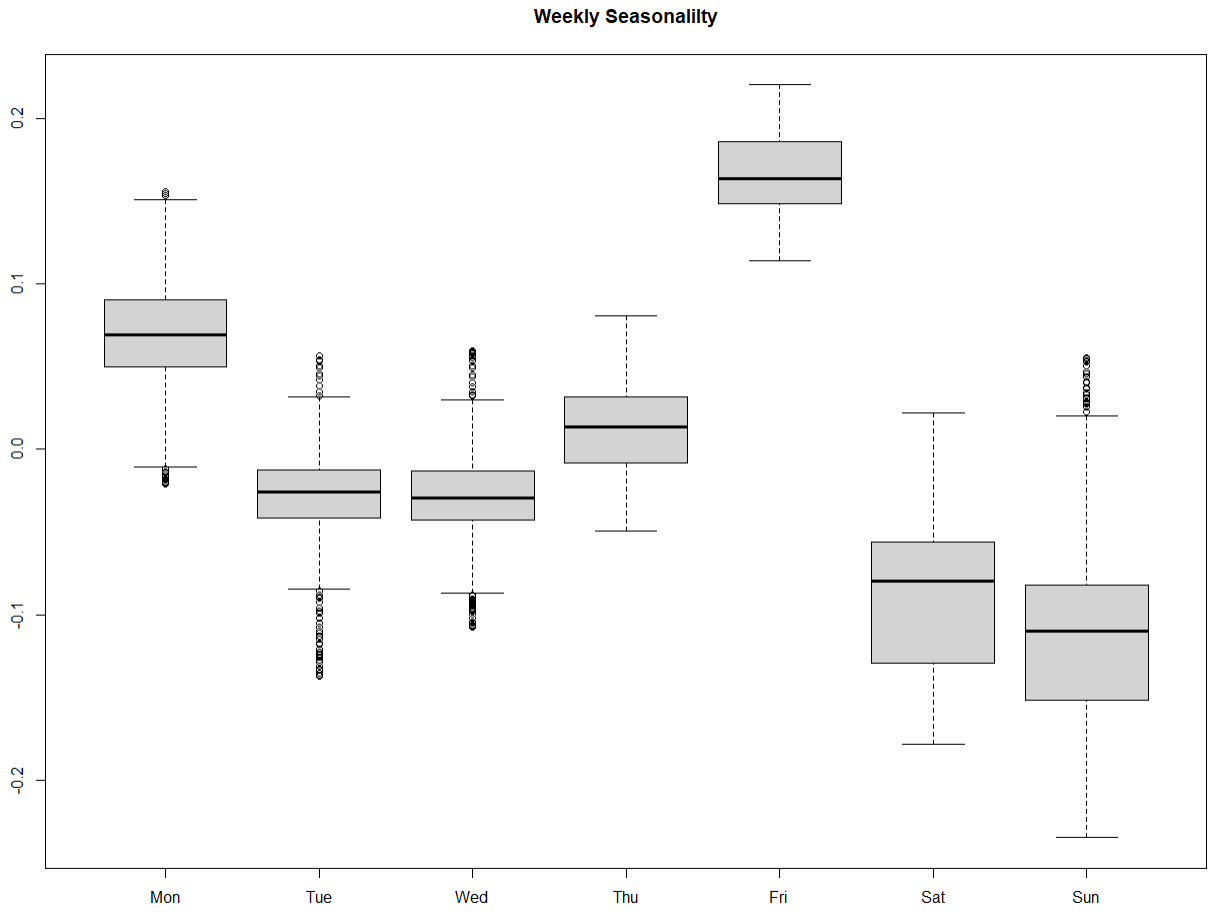
Weekly Seas.

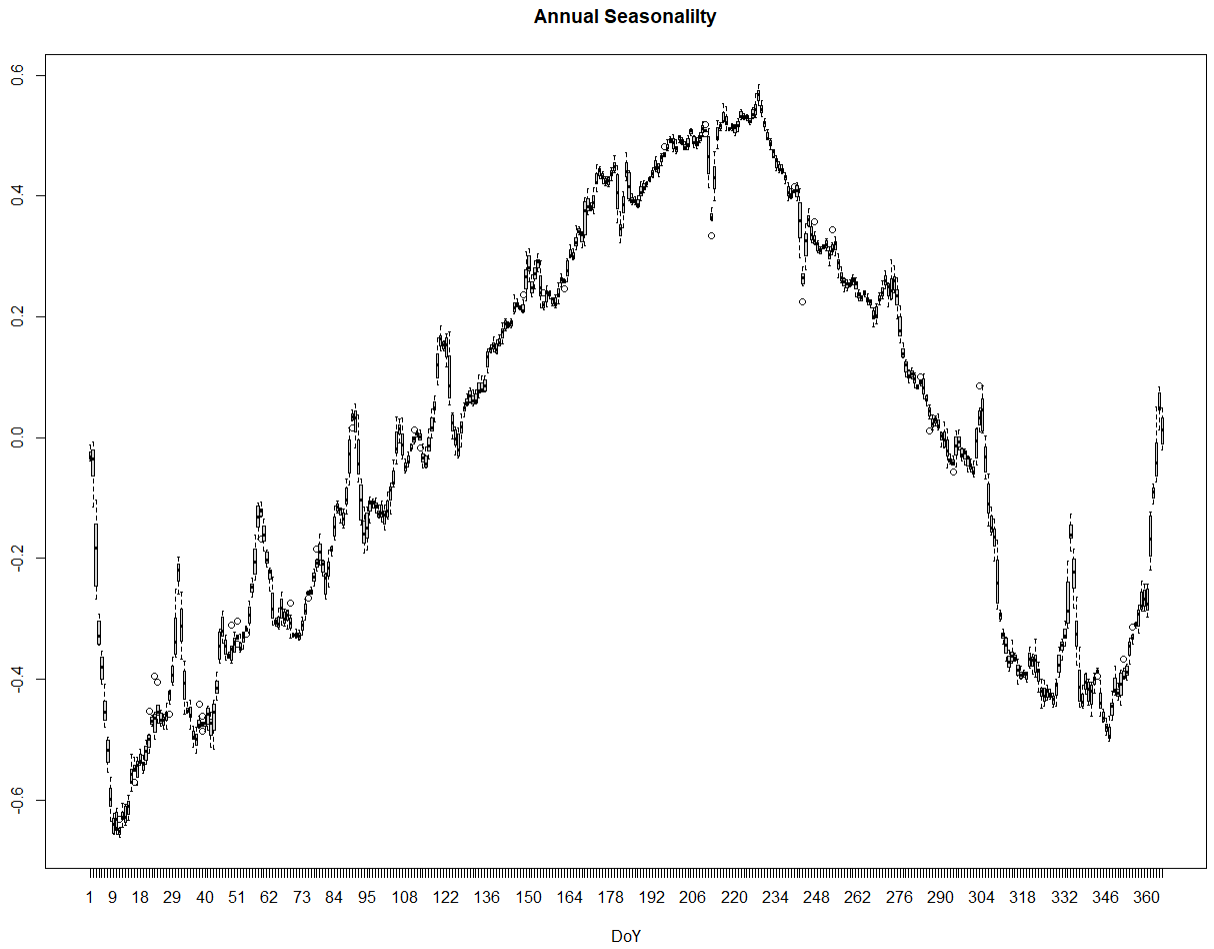


Weekly Seas.

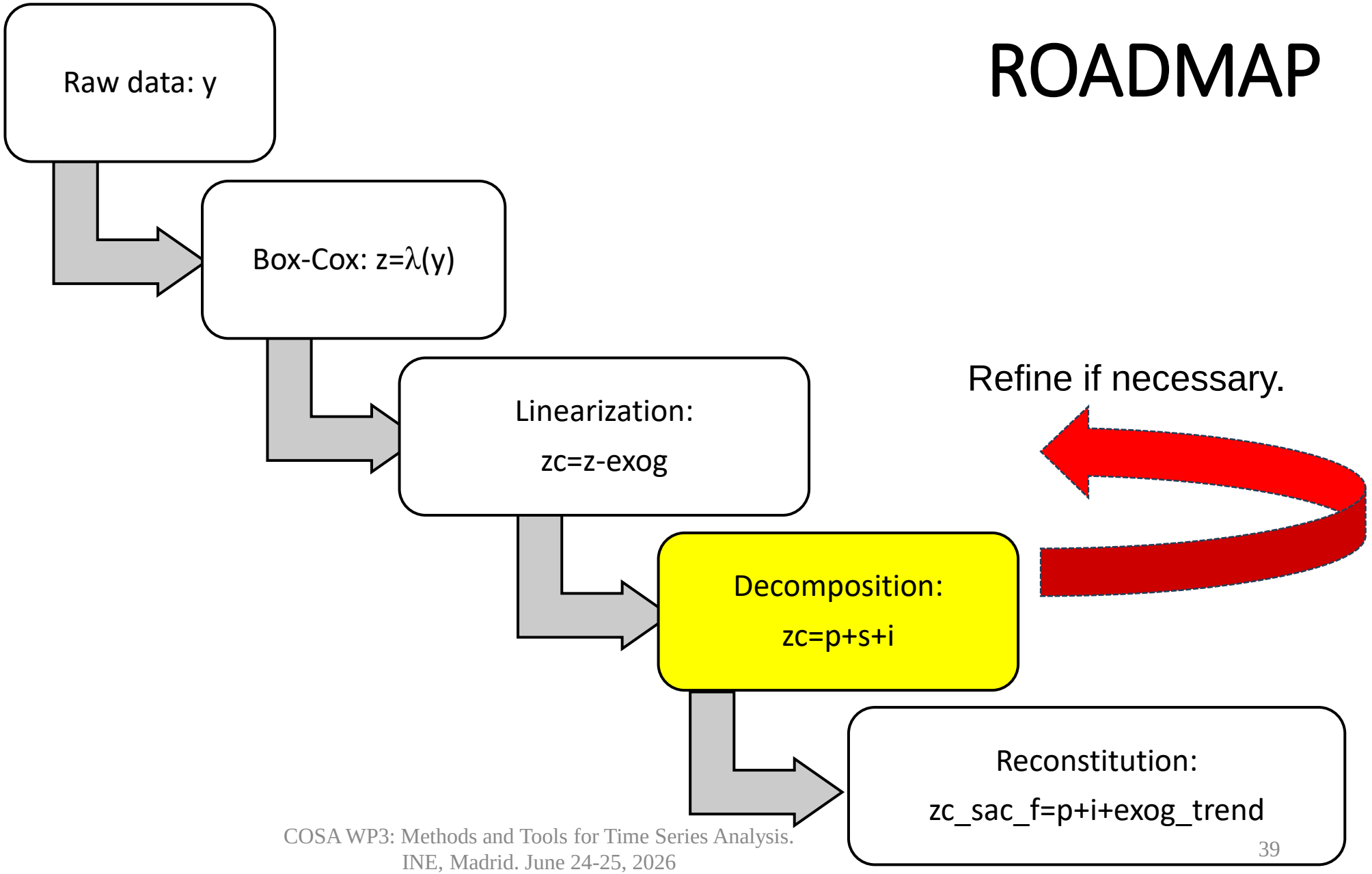


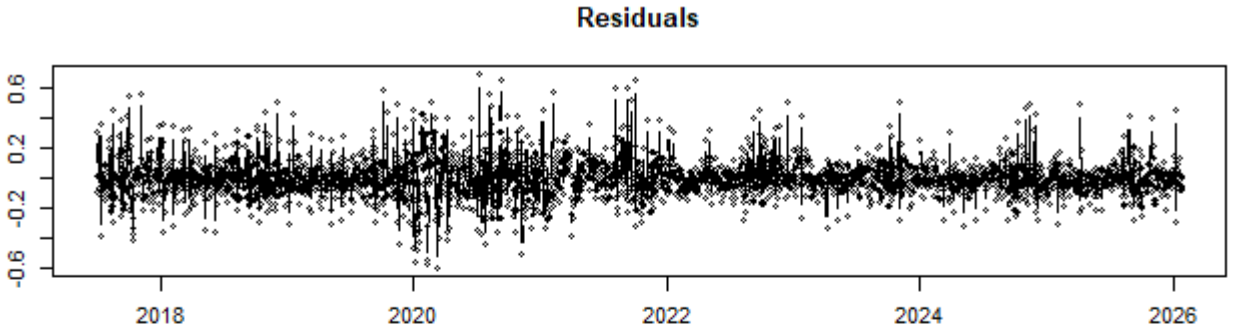

Annual Seas.
0.4
0.0
-0.4
2018
2020
2022
2024
2026

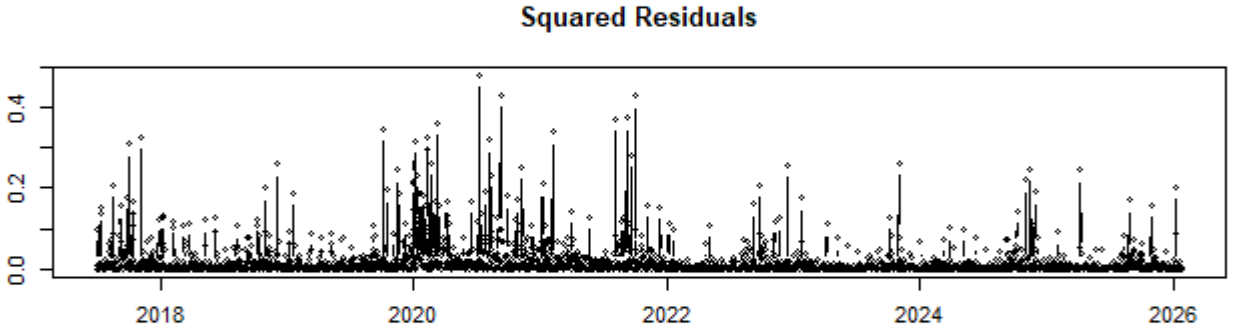

Weekly Seasonalilty
0.2
0.1
0.0
-0.1
-0.2
Mon
Tue
Wed
Thu
Fri
Sat
Sun

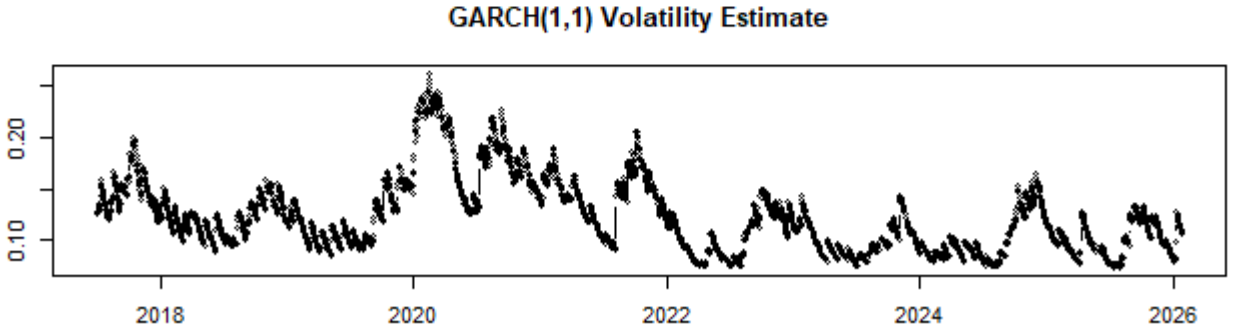

Annual Seasonalilty
0.6
0.4
0.2
0.0
-0.2
-0.4
-0.6
1 9 18 29 40 51 62 73 84 95 108 122 136 150 164 178 192 206 220 234 248 262 276 290 304 318 332 346 360
DoY

ROADMAP

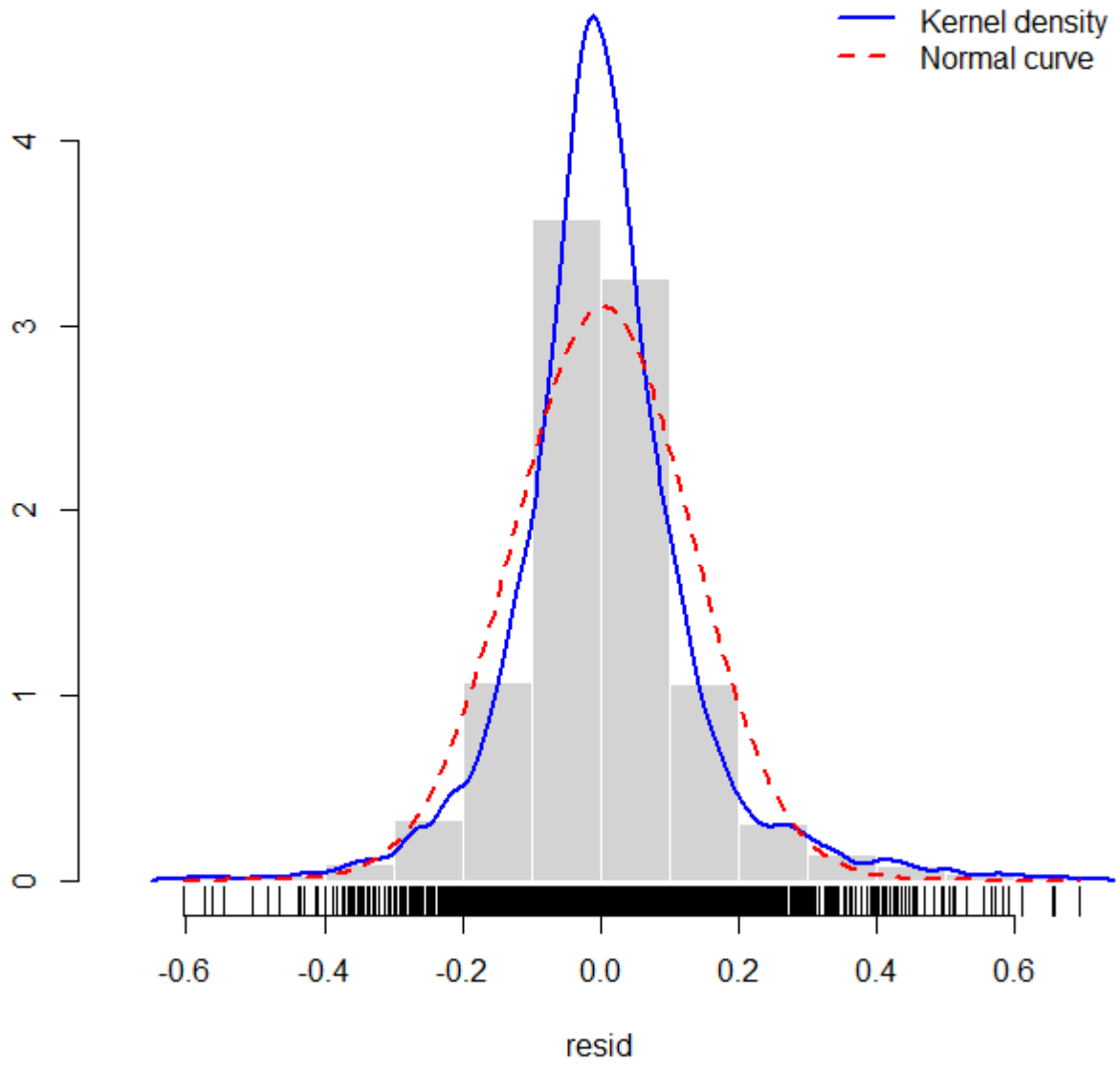

Raw data: y
Box-Cox: z=λ(y)
Linearization:
zc=z-exog
Decomposition:
zc=p+s+i
Reconstitution:
zc_sac_f=p+i+exog_trend
Refine if necessary.



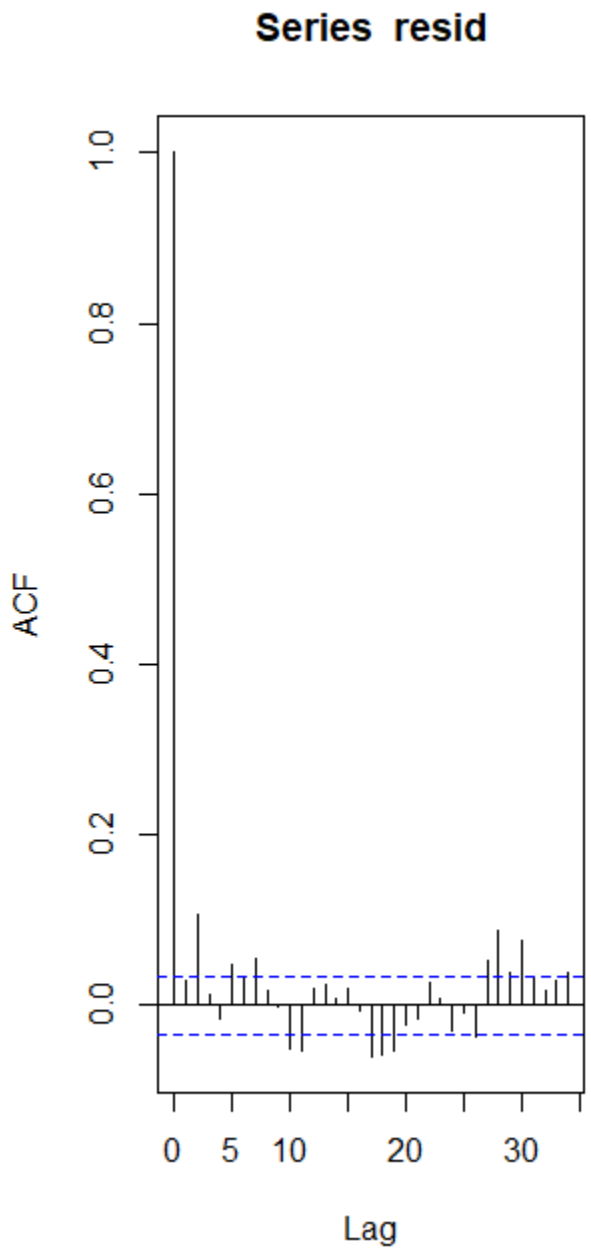


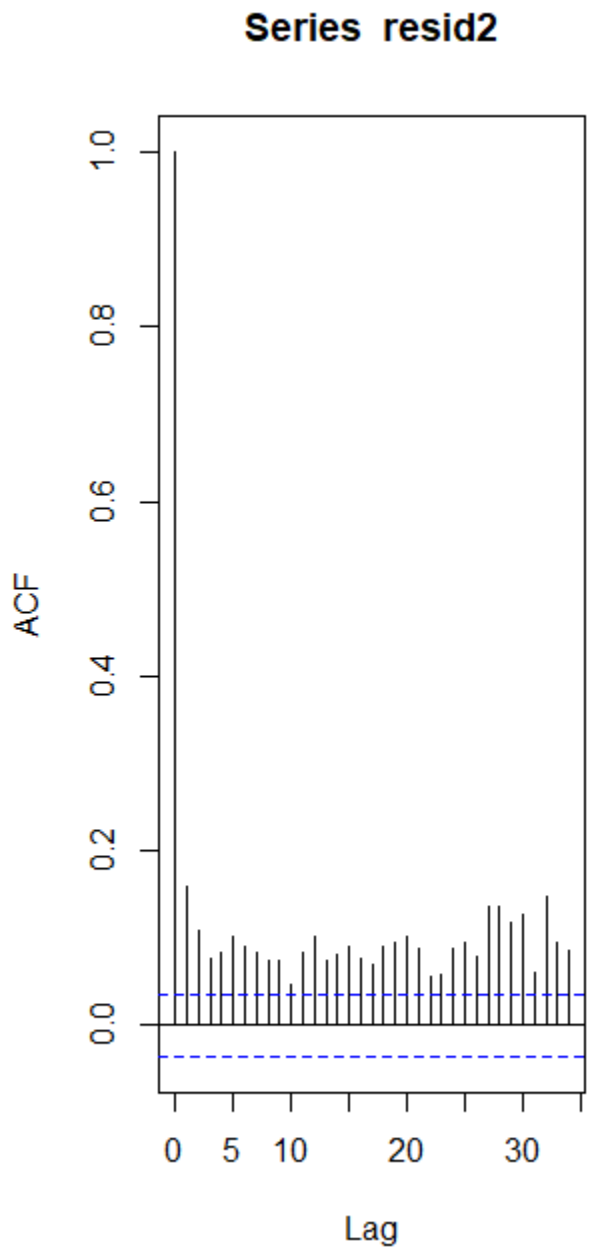


- Some inhomogeneities.
- Mainly linked to the CV19 crisis.
- Volatility clustering, even after outliers correction.



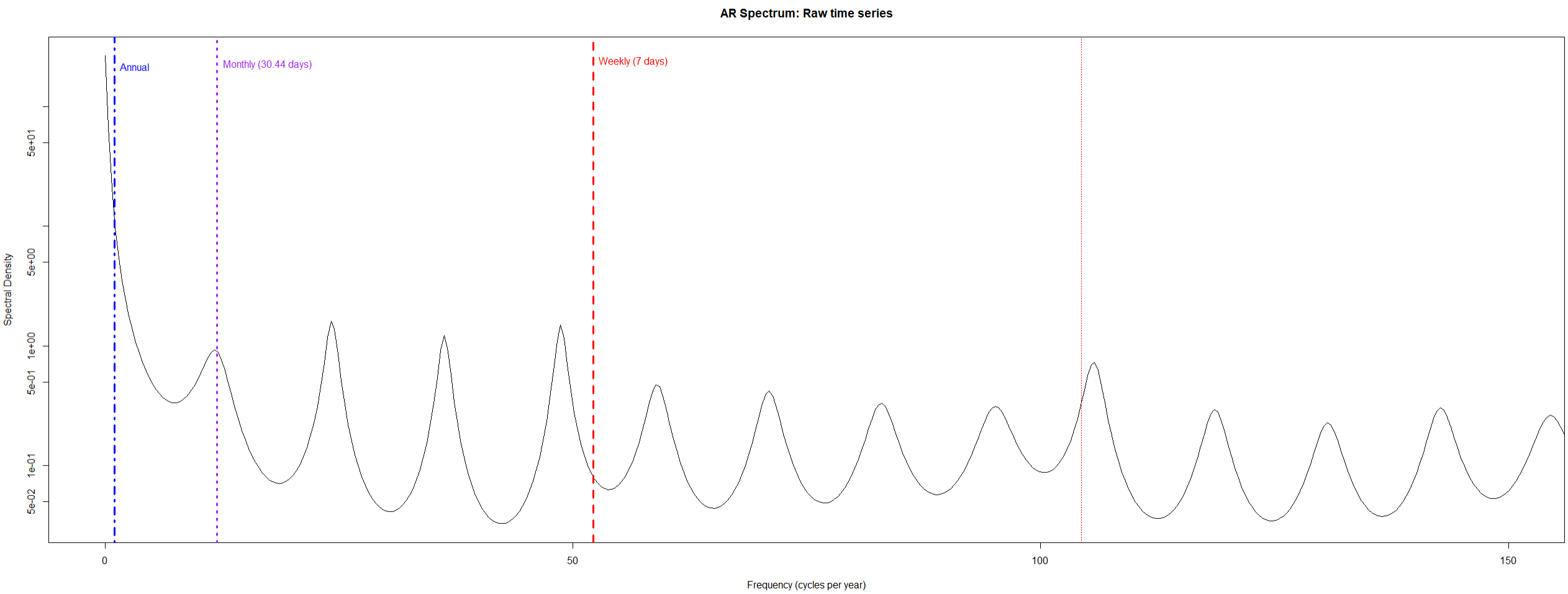
- Excessive kurtosis.
- Slight positive asymmetry.
- A clear departure from the Gaussian case.
- Possibly due to the CV19-induced inhomogeneities.
- (See Appendix “CV19 Shock”).



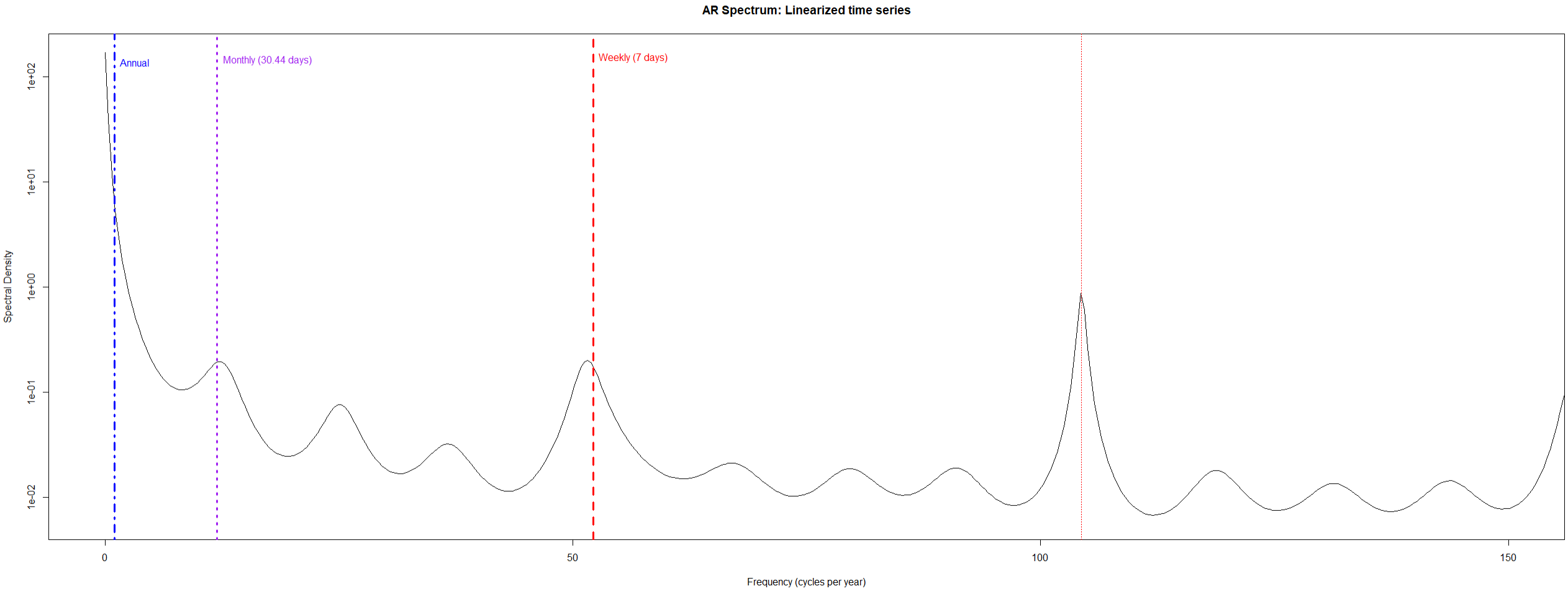


- An acceptable lack of serial dependence…
- But a clear evidence of heteroskedasticity.

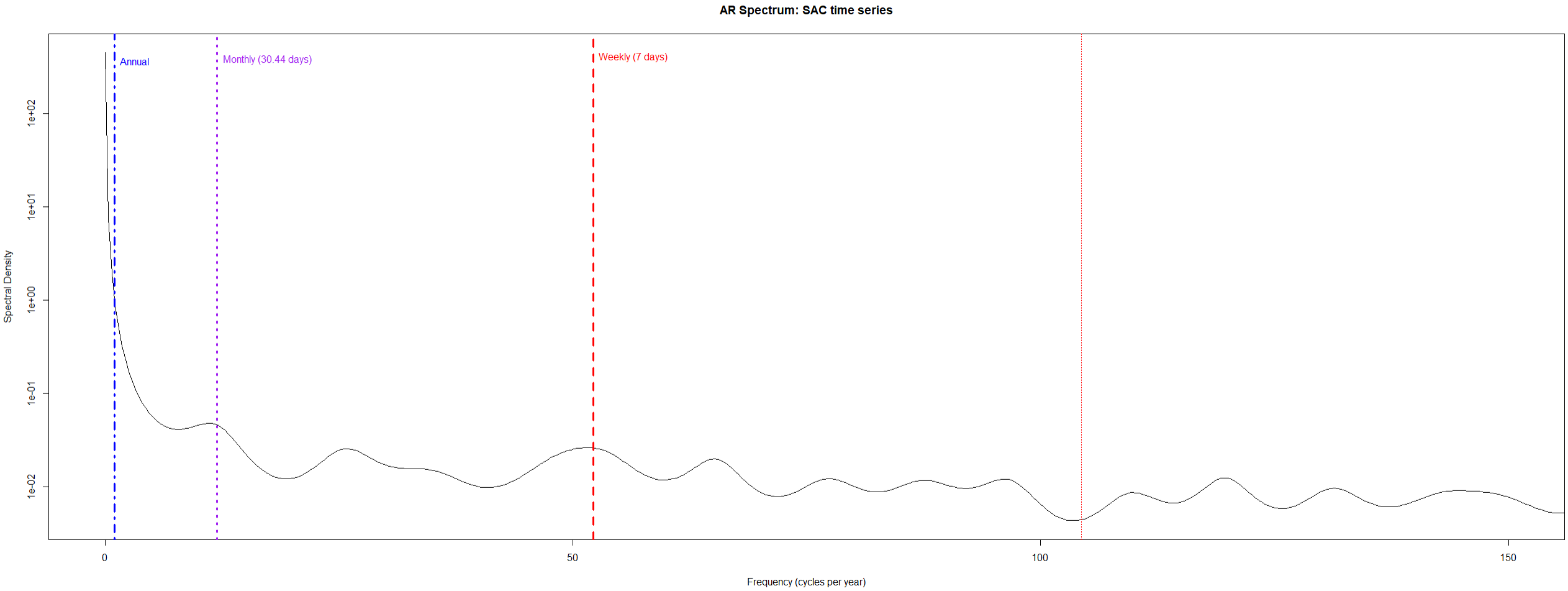
Raw (log) data


AR Spectrum: Raw time series
Annual
Monthly (30.44 days)
Weekly (7 days)
Spectral Density
5e-02
1e-01
5e-01
1e+00
5e+00
5e+01
0
50
100
150
Frequency (cycles per year)

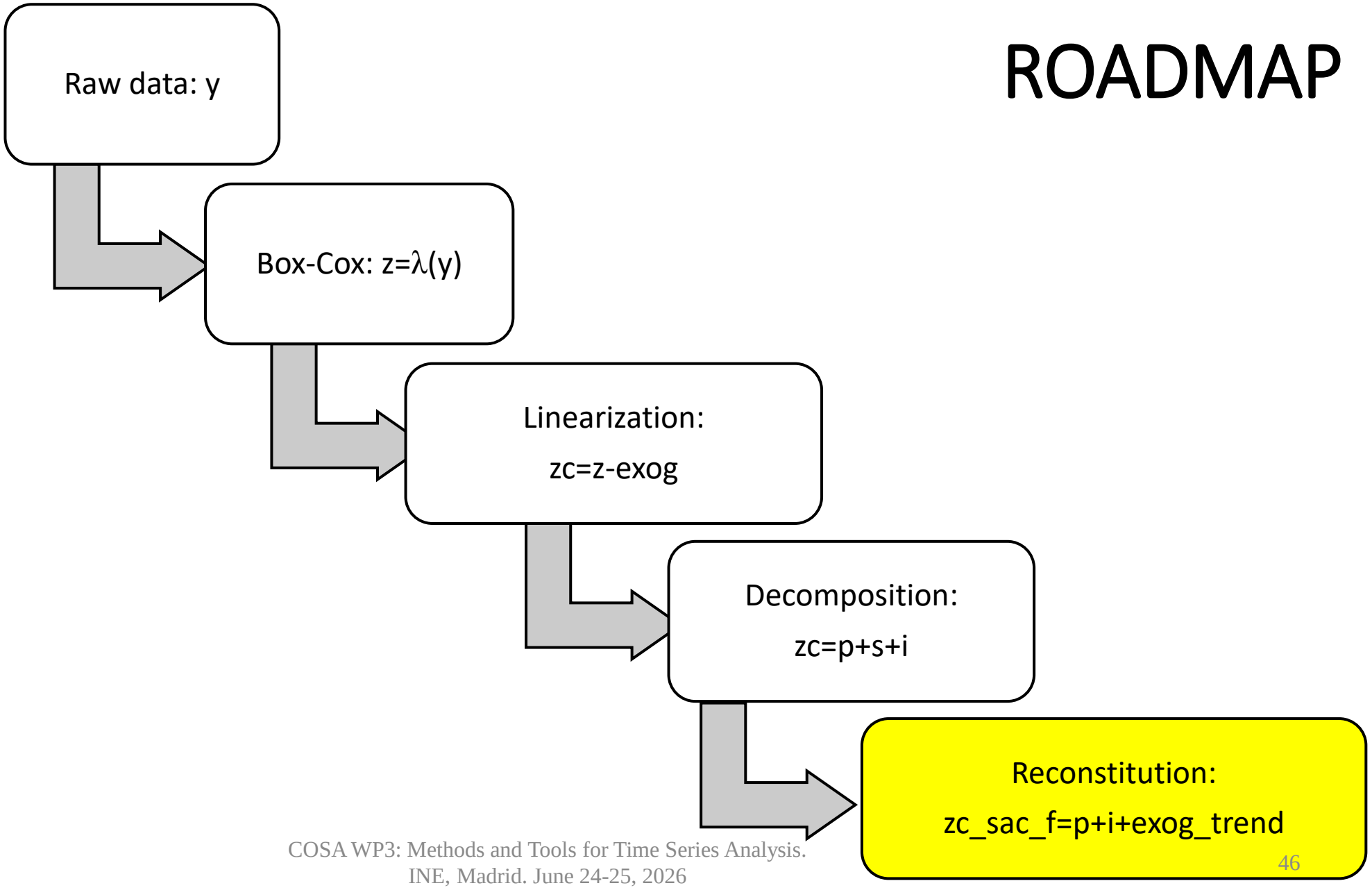
Linearized data

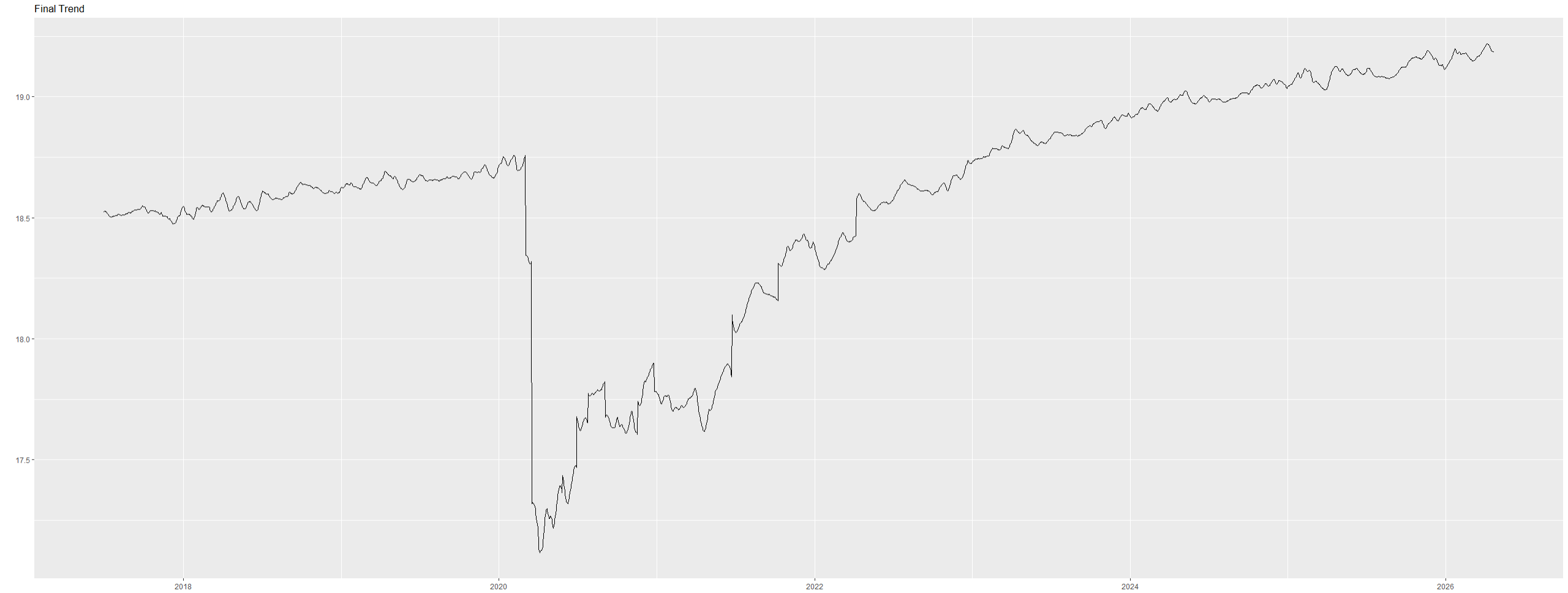


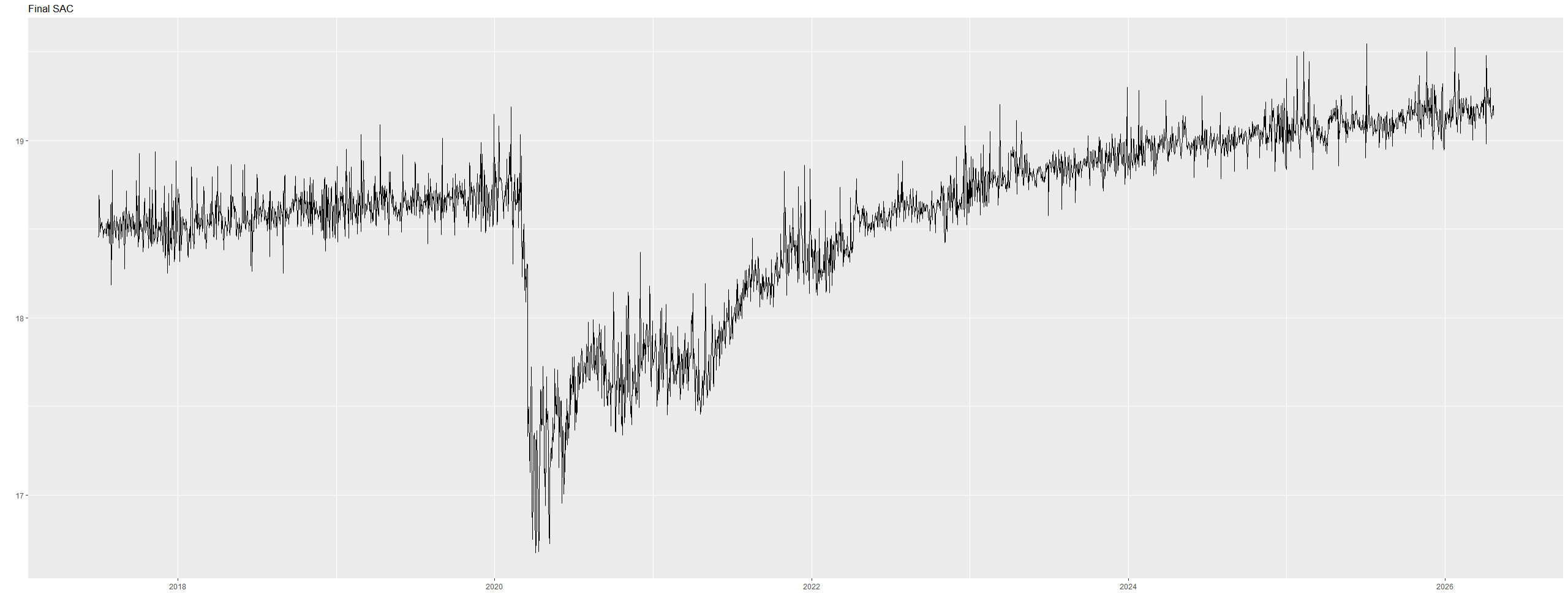
Seasonally and calendar adjusted data



ROADMAP


Raw data: y
Box-Cox: z=λ(y)
Linearization:
zc=z-exog
Decomposition:
zc=p+s+i
Reconstitution:
zc_sac_f=p+i+exog_trend


Final Trend
19.0
18.5
18.0
17.5
2018
2020
2022
2024
2026


Final SAC
19
18
17
2018
2020
2022
2024
2026

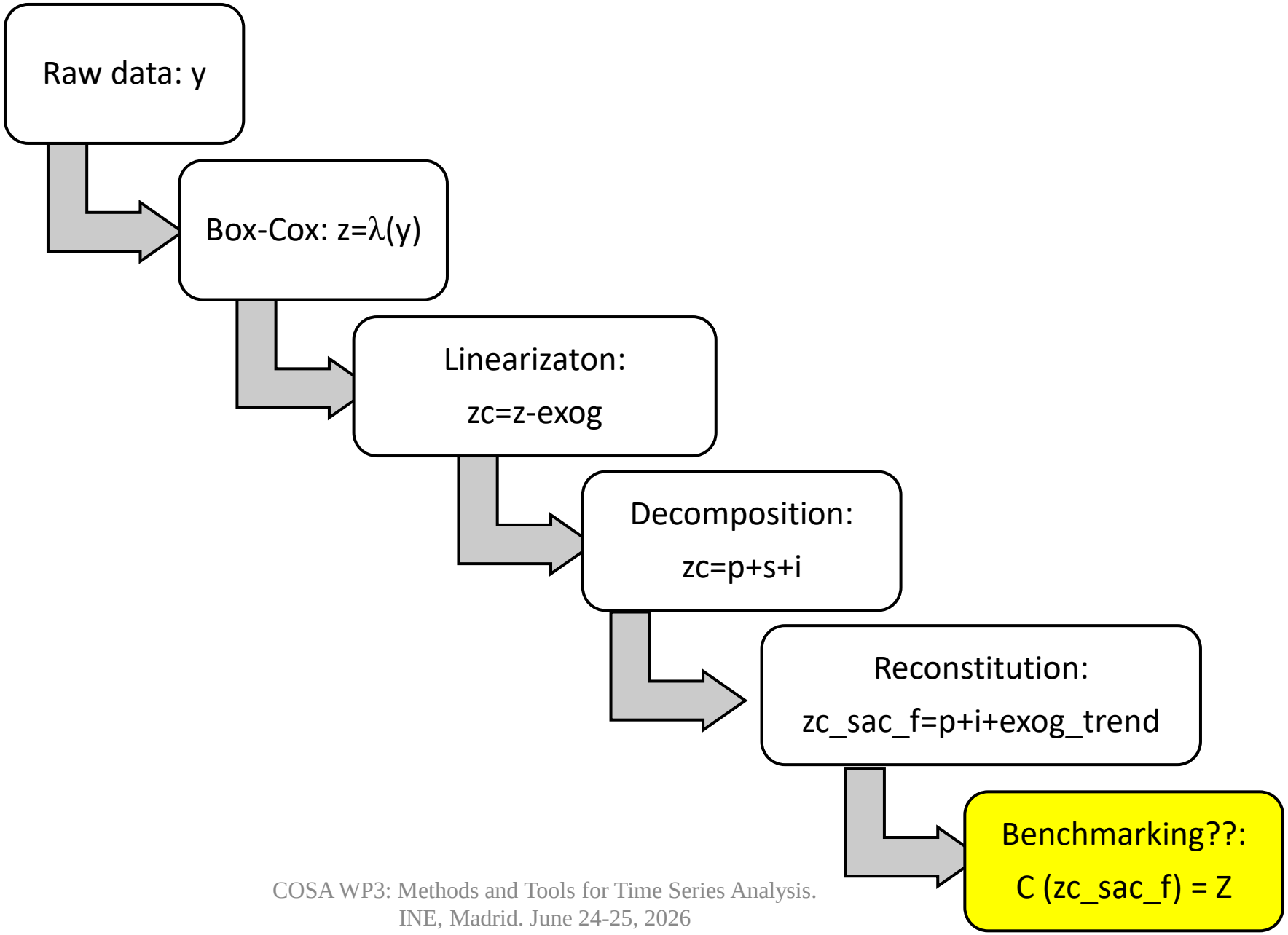
CONCLUSIONS AND FUTURE STEPS

CONCLUSIONS

- AMB modeling and decomposition provide a sensible way to deal with real economic (multi) seasonal time series.
- Pre-processing is critical, especially to deal with “spiky” monthly patterns and CV19 outliers.
- Subsampling helps to explore the main features of the data (e.g. number and periodicity of seasonals).
- Diagnosis is greatly improved by the availability of AMB residuals ➔ a great advantage of the AMB approach.
- Spectral analysis provides a simple and informative diagnosis.

FUTURE STEPS

- Some pre-testing for the Box-Cox transformation may be useful, even if only the polar cases (log/no-log) are considered.
- How to improve the modeling of monthly patterns? Time-varying effects?
- Incorporation of the seasonality tests as provided by the `seastests` R package (Ollech and Webel, 2025a, 2025b).
- Benchmarking to monthly totals? If yes, how? An adapted Denton procedure?
- Forecasting performance for nowcasting.


Raw data: y
Box-Cox: z=λ(y)
Linearizaton:
zc=z-exog
Decomposition:
zc=p+s+i
Reconstitution:
zc_sac_f=p+i+exog_trend
Benchmarking??:
C (zc_sac_f) = Z

THANKS FOR YOUR ATTENTION!!

REFERENCES

- Guerrero, V. (1993). Time-series Analysis Supported by Power Transformations, Journal of Forecasting, 12, 37-48.
- Ollech, D. and Webel, K. (2025a). Package `seastests`. CRAN Repository.
- Ollech, D. and Webel, K. (2025b). A Novel Tree-Based Combined Test for Seasonality, Data Science in Science, 4:1, 1-11.

APPENDIXES

HORECA: NACE-25 Components

- 5510 – Hotels and similar accommodation.
- 5520 – Holiday accommodation and other short-stay lodging.
- 5530 – Campsites and caravan parks.
- 5590 – Other accommodation.
- 5610 – Restaurants and mobile food service activities.
- 5621 – Event catering (provision of prepared meals for events).
- 5629 – Other food service activities.
- 5630 – Beverage-serving establishments (bars, cafes, pubs)

CALENDAR EFFECTS

- Exogenous dummy variables generated by an specific function that receives as input a table with the corresponding dates.

```
# Loading impulse dates from external file (yy-mm-dd)
D_tau <- import('impulse_dates.xlsx', which = 'TAU')

# Applying function
X <- impulse_var(d_ini,
                 d_fin,
                 D_tau)
```

C18

	A	B	C	D
1	year	month	day	
2	2021	10	12	
3	2021	11	9	
4	2021	12	6	
5	2021	12	8	
6	2022	10	12	
7	2022	11	9	
8	2022	12	6	
9	2022	12	8	

- Alternatively, we can use the functionalities provided by `rjd3toolkit` to generate calendars (national calendars, holidays, etc.).

CALENDAR EFFECTS

	NATIONAL HOLIDAYS: Reference date																SPECIAL DAYS: Reference date			
	January 1		January 6		May 1		August 15		October 12		November 1		December 6		December 8		December 25		December 24		December 31	
	NEW YEAR		FEAST OF THE EPIPHANY		LABOR DAY		ASSUMPTION OF THE VIRGIN		NATIONAL DAY		ALL SAINTS		CONSTITUTION DAY		IMMACULATE CONCEPTION		CHRISTMAS		CHRISTMAS EVE		NEW YEAR'S EVE	
2017	1	1	1	6	5	1	8	15	10	12	11	1	12	6	12	8	12	25	12	24	12	31
2018	1	2	1	6	5	1	8	15	10	12	11	1	12	6	12	8	12	25	12	24	12	31
2019	1	1	1	7	5	1	8	15	10	12	11	1	12	6	12	9	12	25	12	24	12	31
2020	1	1	1	6	5	1	8	15	10	12	11	2	12	7	12	8	12	25	12	24	12	31
2021	1	1	1	6	5	1	8	16	10	12	11	1	12	6	12	9	12	25	12	24	12	31
2022	1	1	1	6	5	2	8	15	10	12	11	1	12	6	12	8	12	25	12	24	12	31
2023	1	2	1	6	5	1	8	15	10	12	11	1	12	6	12	8	12	25	12	24	12	31
2024	1	1	1	6	5	1	8	15	10	12	11	1	12	6	12	9	12	25	12	24	12	31
2025	1	1	1	6	5	1	8	15	10	13	11	1	12	6	12	8	12	25	12	24	12	31
2026	1	1	1	6	5	1	8	15	10	12	11	2	12	7	12	8	12	25	12	24	12	31

CALENDAR EFFECTS

	EASTER WEEK									
	GOOD THURSDAY (Regional holiday)		GOOD THURSDAY (National holiday)		GOOD THURSDAY		EASTER		EASTER MONDAY (Regional holiday)	
2017	4	13	4	14	4	15	4	16	4	17
2018	3	29	3	30	3	31	4	1	4	1
2019	4	18	4	19	4	20	4	21	4	22
2020	4	9	4	10	4	11	4	12	4	13
2021	4	1	4	2	4	3	4	4	4	5
2022	4	14	4	15	4	16	4	17	4	18
2023	4	6	4	7	4	8	4	9	4	10
2024	3	28	3	29	3	30	3	31	4	1
2025	4	17	4	18	4	19	4	20	4	21
2026	4	2	4	3	4	4	4	5	4	6

CALENDAR EFFECTS

	"BIG" REGIONAL HOLIDAYS											
	February 28		September 11		June 24		December 26		May 2		Movable	
	AND		CAT		CAT1		CAT2		MAD		MAD2	
2017	2	28	9	11	6	24	12	26	5	2	3	20
2018	2	28	9	11	NA	NA	12	26	5	2	NA	NA
2019	2	28	9	11	6	24	12	26	5	2	NA	NA
2020	2	28	9	11	6	24	12	26	5	2	NA	NA
2021	3	1	9	11	6	24	NA	NA	5	3	3	19
2022	2	28	9	11	6	24	12	26	5	2	7	25
2023	2	28	9	11	6	24	12	26	5	2	3	20
2024	2	28	9	11	6	24	12	26	5	2	7	25
2025	2	28	9	11	6	24	12	26	5	2	7	25
2026	2	28	9	11	6	24	12	26	5	2	NA	NA

MONTHLY SEASONALITY

- **Robert Hyndman's blog**

- **Monthly seasonality**
  3 May 2017

I regularly get asked why I don't consider monthly seasonality in my models for daily or sub-daily time series. For example, this recent comment on my post on seasonal periods, or this comment on my post on daily data. The fact is, I've never seen a time series with monthly seasonality, although that does not mean it does not exist.

Monthly seasonality is defined as a regular pattern that recurs every month in data that is observed more frequently than monthly. This is different from the seasonality you frequently see in monthly observations which exhibit an annual seasonality.

Monthly seasonality would occur if there is some regular activity that takes place every month and which affects the time series. For example, some companies try to average their expenditure across the month and often have to spend more at the end of the month to justify the budget. So daily expenditure tends to increase at the end of each month, producing a monthly seasonal pattern.

Or imagine a situation where a company always stocks up on supplies on the second Tuesday in every month. Then daily expenditure would have a monthly seasonal pattern.

This type of seasonal pattern is quite difficult to model. The major problem is that months are not of equal length, and Tuesdays do not always fall on the same day of the month. So unlike weekly and yearly seasonality, the seasonal pattern is not periodic and cannot be modelled in the same way that we model other types of seasonality.

The best approach is to consider what is causing the monthly seasonality, and try to allow for it explicitly using regressor variables. If there is an end-of-month effect, then a dummy variable indicating which day(s) are at the end of the month would effectively handle it. If there is a regular event on a specific day in the month (such as the second Tuesday), then again a dummy variable would be best used.

Approximating monthly seasonality by assuming the months are all of 30 days is unlikely to be effective when the seasonality is induced by these types of business practices.

CV19 SHOCK

- A simple comparison between a prior intervention, based on the official States of Alarm (three dummies), and an empirical estimate based on the LS outliers detected by AMB after March 1st 2020.
- The dates for the intervention are:

State of Alarm	Start	End	Duration (days)
1	14/03/2020	21/06/2020	100
2 (Partial)	09/10/2020	24/10/2020	16
3	25/10/2020	09/05/2021	197

CV19 SHOCK

- A striking difference between prior and empirics:
  - Much larger cumulated impact: ~2,17 times.
  - Much larger duration: ~1y more.

